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RESEARCH-ARTICLE

Remote Breathing Rate Tracking in Stationary Position Using the Motion and Acoustic Sensors of Earables

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ABSTRACT

Breathing rate is critical for the user's respiratory health and is hard to track outside the clinical context, requiring specialized devices. Earables could provide a convenient solution to track the breathing rate anywhere by leveraging the user's breathing-related motion and sound captured through the earables' motion sensors and microphones. However, small non-breathing head movements or background noises during the assessment affect the estimation accuracy. While noise filtering improves accuracy, it can discard valid measurements. This paper presents a multimodal approach to tracking the user's breathing rate using a signal-processing-based algorithm on motion sensors and a lightweight machine-learning algorithm on acoustic sensors from the earables that balances the accuracy and data retention. A user study with 30 participants shows that the system can accurately calculate breathing rate (Mean Absolute Error < 2 breaths per minute) while retaining most breathing sessions (75% performed in real-world settings). This work provides an essential direction for remote breathing rate monitoring.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing design and evaluation methods**; Empirical studies in ubiquitous and mobile computing.

KEYWORDS

Breathing, Hearable, Breathing Rate, Remote Monitoring

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1 INTRODUCTION

Breathing is fundamental to our health and wellness, and the number of breaths we take per minute (breathing rate) is a critical vital sign. A healthy adult takes 12-20 breaths per minute (BPM), and abnormalities in resting breathing rate are associated with severe medical conditions, including cardiac arrest, pneumonia, and respiratory infections [17, 46, 55]. Breathing rate is a highly desirable feature for remote condition monitoring for people with respiratory diseases such as Chronic Obstructive Pulmonary Disease (COPD) and Asthma [60, 89], as it could lead to the early identification of disease exacerbation [87]. In addition to acute conditions, breathing rate is also a good indicator of fitness level [53–55], mental stress [25, 44, 74, 79], and cognitive load [22]. Therefore, daily monitoring of breathing rates could offer significant health benefits to everyone. However, the existing approaches to actively tracking the breathing rate are not very convenient. People can track their breathing rate by manually counting the number of breaths per minute or using an expensive device like chestband [76].

Realizing the need for a more convenient solution, researchers explored mobile and wearable device-based solutions for remote breathing rate monitoring. Many works suggest that it is feasible to calculate the breathing rate by utilizing the motion and acoustic sensors from smartphone [3, 60, 63], smartwatch [27, 29, 39, 75], and earables [2, 37, 62, 67]. While existing works can detect the breathing rate with good accuracy, both motion and acoustic sensors have limitations. Breathing produces an extremely subtle body motion and sound; small movements or background noises can easily mask them. Differentiating the breathing signals from the body movements and background noises is challenging; therefore, most works discard the noisy signals by using signal processing and machine learning algorithm. While discarding noisy signals is critical for estimating the breathing rate with high accuracy, it negatively impacts the session retention rate¹ [14, 39]. For example, even in the controlled resting position, the smartwatch-based approach can retain only 7–16.5% windows [39]. While accuracy is critical for any health monitoring, dropping a large number of sessions could reduce the usability of the system. Since both motion and acoustic sensing-based approaches can be inaccurate in different circumstances, one potential way to improve the retention rate could be using a multimodal solution where the system can

¹Session Retention Rate = $\frac{\text{No. of the retained session after filtering}}{\text{total breathing sessions}}$. Each breathing rate assessment can be considered a breathing session.

leverage both sensors. For example, when breathing rate estimation is inaccurate using the motion sensors due to body movements, the system can utilize audio to assess the breathing rate. However, multimodal sensing is not feasible on many commercially available popular wearable devices due to the inconvenient placement of the devices. For example, smartwatches are placed far away from our nose or mouth, where breathing sounds are produced.

This work presents a multimodal approach to tracking the breathing rate using an ‘earable’² [11], as they can provide a convenient method for monitoring. Earables are already a well-established form factor [11] and people spend 3–4 hours per day on the earables [84]. More importantly, they are very well-suited for capturing breathing motions and sounds. On the one hand, earables can pick the subtle vertical head motions produced due to breathing. On the other hand, they can partially hear the breathing sounds generated in our nose and mouth. However, similar to other devices, both audio and motion signals can be easily corrupted by non-breathing movements and sounds. Therefore, a multimodal approach using the motion and acoustic sensor could be more applicable for earbuds to estimate the breathing rate with a high retention rate. Most existing works [37, 63, 67] that used earbuds for breathing rate monitoring use a single sensor in a supervised setting; a multimodal approach using the earbuds is yet to be explored.

To develop and validate a multimodal breathing solution, we conducted a comprehensive study with 30 participants who performed several regular and controlled breathing tasks in a supervised lab and multiple unsupervised home sessions for seven days. Based on collected data, we developed a system that leverages the motion and acoustic sensors of Samsung earbuds to estimate the breathing rate. We use a well-established signal processing-based approach to calculate the breathing rate using the motion sensors. We use a random-forest-based machine learning algorithm for estimating the breathing rate using microphones. When compared with an FDA-approved chestband (Zephyr [76]), our results demonstrate that a multimodal approach can calculate the breathing rate with relatively good accuracy (<2 BPM) while retaining a good percentage of breathing sessions (75.1% sessions). We implemented a partial version of the algorithm on the earbuds, and our analysis suggests that it is feasible to implement it on a low-resource device without consuming significant resources. This work demonstrates the potential to use earbuds for more convenient breathing rate tracking.

The main contribution of this work are listed below:

- We conducted an extensive data collection study using a commercial earbud for breathing rate monitoring. We empirically investigate the challenges of breathing rate tracking using earbuds in the home environment.
- We propose a novel multimodal approach for breathing rate monitoring using a commercial earable. We use a signal processing pipeline for breathing rate monitoring using the motion sensors and a machine learning approach for estimating the breathing rate using acoustic data.
- We evaluated our multimodal approach on the data set collected from both supervised and unsupervised settings. The multimodal approach can estimate the breathing rate with good accuracy and a high retention rate.

²‘Eearable’ to refer to a group of wearable devices that can be worn around our ears [11].

2 BACKGROUND AND RELATED WORK

This section discusses the importance of breathing rate monitoring and some existing approaches for breathing rate tracking.

2.1 Importance of Breathing Rate Monitoring

Breathing rate is a vital sign and significantly impacts people’s health and wellness. Normally, an adult takes 12-20 breaths per minute (BPM) [57, 68]. However, the rate can vary depending on the person’s level of activity [70], age [68], and medical conditions as well as environmental conditions like humidity [71], altitude [12], and temperature [71]. While minor variations in breathing rate do not indicate an emergency, abnormally high or low breathing rate can be an early predictor of acute illness and condition exacerbation. For example, Fieselmann et al. reported that a breathing rate higher than 27 BPM is an important forecaster of cardiac arrest in the hospital [17]. Yanez et al. [87] found that breathing rate can increase during the exacerbation of Chronic Obstructive Pulmonary Disease (COPD). Pain medication can reduce the breathing rate, and a low rate (<8 BPM) could indicate that the patient has taken higher doses of medicine [6]. Breathing rate is also linked with stress, cognitive load, and physical fitness [55]. Stress negatively impacts an individual’s mental well-being [25], and many clinical studies [45, 55, 77, 85] suggest that breathing rate can increase during a stressful event. Ha et al. demonstrated that breathing rate and associated features are important predictors for automated stress monitoring [25]. Breathing rate is also associated with fitness level; Nicolo et al. [54] provided strong evidence that breathing rate is a strong marker of physical effort during a high-intensity exercise.

2.2 Approaches for Breathing Rate Tracking

The most standard approach for breathing rate monitoring relies on manually counting the number of breaths a person takes per minute. Thus, the patient needs to count their breathing or rely on someone else to count their breath. However, the measurement of breathing rate is deemed as a core nursing skill [34], and even experts can make an error in measurements [18]. In addition, self-counting of breathing cycles could result in flawed measurements because once the user is aware of their breathing, they may start breathing faster or slower than usual [50]. An ideal approach to breathing rate counting should passively count the respiratory cycles without interfering with the user’s natural breathing. Clinically, breathing rate can be monitored by a range of techniques, including respiratory belts and impedance pneumography. Pneumography uses electrodes to detect changes in the thoracic electrical impedance for measuring instantaneous airflow changes [72]. Respiratory belts use high-fidelity sensors for passive breathing rate monitoring. For example, Zephyr chestband [76], a U.S. FDA-approved device, uses a pressure sensor to calculate breathing rate by monitoring the rib cage expansion and contraction. Zephyr can estimate the breathing rate with ± 3 BPM accuracy in stationary conditions [76]. However, these devices are expensive and inconvenient for daily usage [41].

2.3 Monitoring Using Ubiquitous Devices

Mobile and wearable devices can provide a more ubiquitous solution for breathing rate monitoring; hence, many recent works propose systems using smartphones, smartwatches, and earables. Table 1

Device	Work	Sensing	Algorithm	Validation Study Description	N	MAE
Phone	Zephyr'16 [3]	IMU	FFT	150 experiments in a controlled setting	7	0.6 [†]
	InstantRR'20 [63]	IMU	ZCR, FFT, Peak	Eight 1-min breathing tasks in a lab	21	0.85
	Nam et al.'15 [49]	Audio	Autoregression	Tracheal and nasal breathing in a office	10	-
	Nam et al. '16 [48]	Camera	Autoregression	Breathing at a metronome rate	11	3.80 [†]
Watch	Wang et al.'21 [81]	Sonar	FFT	3-month study in office, bedroom, & movie theatre	5	0.2 [‡]
	BioWatch'15 [29]	IMU	FFT	Breathing under relaxed & aroused condition in a lab	12	1.27
	WearBreathing'19 [39]	IMU	RF+CNN	~1 hr lab + 3 hrs uncontrolled session	14	1.09
	SleepMonitor'17 [75]	IMU	FFT	70 nights of sleep sessions	16	0.72
	RespWatch'21 [14]	PPG	Peak+CNN	Breathing while performing 4 tasks in a lab	32	2.01
Earables [11]	MindfulWatch'17 [27]	IMU	DTW	36 real-world meditation sessions	11	0.77
	BioGlass [28]	IMU	FFT	Breathing under relaxed & aroused condition in a lab	12	1.18
	Röddiger et al.'19 [67]	IMU	FFT	Breathing in three postures before and after exercise	12	2.55
	Rahman et al.'21 [62]	IMU	ZCR	1.5 min breathing task in a resting condition	30	2.34
	RRMonitor'21 [2]	IMU	ZCR, Peak, FFT	Controlled lab sessions in resting position	30	1.64
	Bestbier & Fourie '18 [5]	PPG	Local Maxima	Controlled trial sessions	16	-0.558 [*]
	Martin & Voix '18 [43]	Audio	Peak	Nose & Mouth Breathing in a sound booth	20	2.7
	Kumar et al.'21 [37]	Audio	LSTM	Workout in both indoor and outdoor environments	21	0.2 [‡]

Table 1: Overview of some existing works that use commercial ubiquitous mobile devices to estimate breathing rate. (MAE = mean absolute error reported as BPM, [†] presented median absolute error, [‡] presented mean squared error, ^{*} presented mean error).

presents a summary of the existing works that leverage various sensors from mobile and wearable devices to estimate the respiration rate. As seen in Table 1, most works utilize inertial measurement units (IMU) from the devices and use an FFT-based algorithm to calculate the respiratory rate. The accuracy of the existing approaches ranges from 0.2–3.80 BPM in terms of mean/median absolute error. We can also observe that most approaches are validated by conducting controlled experiments in constrained settings like a lab or office. However, the challenges of breathing rate monitoring in the home setting are not sufficiently explored. Additionally, most works focused more on the accuracy of breathing rate tracking; few explored the trade-off between accuracy and session retention rate.

Phone-based approaches are not very convenient to the user as the device needs to be placed in a convenient location (e.g., chest or nose), which could be intrusive. Wearable devices could provide a more usable solution, as they don't interrupt the users' natural interaction with the device. As can be seen in Table 1, many previous works used watches and earables for estimating the breathing rate. While MindfulWatch [27], SleepMonitor [75], and Kumar *et al.* [37] achieved impressive performance, they evaluated their approach in extreme cases (sleep [75], meditation [27], and exercise-induced breathing [37]), which does not resemble the user's natural breathing scenarios in a stationary position. In natural breathing scenarios, it is expected that the user will have more motion artifacts and background noises, which can drastically reduce the accuracy. Most works discard the noisy signals by employing signal processing or machine learning techniques, which improves the accuracy but discards the breathing sessions. For example, WearBreathing [39] achieved impressive accuracy (MAE 1.09 BPM) with only 7–16.5% data retention in a resting position (sitting, standing, and lying). Such a low retention rate is not usable to the user, as they cannot make successful measurements in most attempts. Therefore, we need a more usable solution that estimates the breathing rate with reasonably good accuracy and retention rate.

Due to the rising popularity of earable devices, several recent works used earables to estimate the breathing rate (Table 1). Both Röddiger *et al.* [66] and Ne *et al.* [51] reviewed the existing works in breathing rate using earables. As seen in Table 1, most works

utilized a single sensor (motion, acoustic, or ppg) from the earbuds to estimate the breathing rates. Unlike wearables, earables demonstrate the potential of using acoustic sensors to estimate breathing rates. However, while most works were conducted in a controlled setting, the accuracy of respiratory rate estimation using the motion and acoustic sensors seems to be not as good as the wearables. People tend to move their heads more frequently than their wrists; therefore, there is a higher chance of motion artifacts in the head region. The audio captured by earables also seems poor quality due to the distance from nose to ear [21, 51]. Hence, the weak performance of motion sensors can be attributed to the low-quality signal captured by earables. While filtering the low-quality signal can improve the accuracy, the data window retention rate will be lower, impacting the system's utility. Therefore, in this work, we use a multimodal approach using a commercially available earable device to explore the trade-off between accuracy and data retention. We extensively evaluated the multimodal approach with a data collection study from the lab and home environments.

3 DATA COLLECTION AND PREPARATION

We conducted a comprehensive data collection study for seven days with 30 participants in *Lab* and *Home* environments using Samsung earbuds. In this section, we discuss our data collection and annotation procedures.

3.1 Data Collection

We use a consulting firm specializing in human factors and usability studies for the participant recruitment and management of the study. Our legal team and the consulting firm's institutional review board (IRB) approved the study. The study was conducted during the COVID-19 pandemic (Feb-Mar, 2021), and we followed a rigorous safety protocol for the study. For example, participants and researchers wore face masks and kept a six-feet distance, repeated cleaning of the surface, and we did not reuse the devices.

3.1.1 Devices & Sensors. We collected the audio (internal microphone) and 6-axis inertial measurement unit (IMU) data using a Samsung Galaxy Buds Pro. The audios were recorded at a 16KHz sampling rate, and IMU data was collected at a 48Hz sampling rate.

Settings	Task Information							Annotable Sessions
	Postures	#Seq	TaskName	Details	N	Length	Task Frequency	
Lab		1	Nose Breathing	Breathing through nose	30	1.5 min	Once	30
		2	Mouth Breathing	Breathing through mouth	30	1.5 min	Once	28
		3	Head Motion	Breathing with head motion	30	2 min	Once	17
		4	Music	Music played on earbuds while breathing naturally	30	1 min	Once	28
	Sitting	5	Regular Breathing	Natural Breathing	30	1 min	Once	30
		9	Cool Down	Recovery breathing after 6-minute walk	30	1 min	Once	30
	Standing	6	Nose Breathing	Breathing through nose	30	1 min	Once	30
	Lying	7	Nose Breathing	Breathing through nose	30	1 min	Once	28
Home		8	Mouth Breathing	Breathing through mouth	30	1 min	Once	30
	Sitting	1	Regular Breathing	Natural Breathing	452	1 min	2x per day	324
		2	Music	Music played on earbuds while breathing naturally	445	1min	2x per day	326
		4	Slow Breathing	Breathing exercise performing 5 breaths per minute	423	1 min	2x per day	205
		x	Nose Breathing	Breathing through nose	141	1 min	voluntary	112
		x	Mouth Breathing	Breathing through the mouth	118	1 min	voluntary	95
	Standing	3a.	Regular Breathing	Natural Breathing	234	1 min	2x on alternate days	175
		x	Nose Breathing	Breathing through nose	119	1 min	voluntary	90
		3b.	Regular Breathing	Natural Breathing	221	1 min	2x on alternate days	178
	Lying	x	Nose Breathing	Breathing through nose	112	1 min	voluntary	93
		x	Mouth Breathing	Breathing through mouth	100	1 min	voluntary	92

Table 2: List of breathing tasks in the study protocol. Unless specified the breathing types, the participants performed their natural breathing during most tasks.

We collected the ground truth breathing rate using a Zephyr Chestband at an 18Hz sampling rate. Zephyr is a Bluetooth-connected FDA-approved chestband that uses a pressure sensor to detect chest expansion and contraction for estimating the breathing rate [76]. The sensors were paired with a Samsung Galaxy S9 running on Android 10 for local data storage. We developed an Android application for data collection. The app periodically transferred the data to a HIPAA-compliant Amazon Web Storage.

3.1.2 Data Collection Protocol. The participants were required to participate in a 45-min lab session and two daily data-collection sessions for seven days at their homes during the participation. The participants can also voluntarily perform several breathing sessions as many times as they prefer at home. At the beginning of the study, the lab sessions were conducted in a controlled environment (with minimal environmental noises and no head motion) for the participants' baseline data collection and training. At the end of the lab sessions, we gave all devices to our participants to take with them to their residences for In-Home sessions. We conducted two surveys at the beginning and end of the study, asking the participants about their demographics and device usage.

In-Lab Session Protocol. During the lab sessions, the participants were required to perform breathing tasks in different postures, activities, and types. The task list and sequence of the lab sessions are listed in Table 2. Two researchers conducted the sessions using the Android app. The researchers followed the task order displayed on the study app, and a timer tracked the task timings. We use the task timings to segment and synchronize the data. The lab sessions started with 1.5 mins of nose and mouth breathing, where we asked our participants to breathe through their nose or mouth. After the nose and mouth breathing tasks, the participants performed head motion tasks with up-to-down (Figure 14a), left-to-right tilting

(Figure 14b), and left-to-right rotation (Figure 14c) movements for 2mins. After the head motion task, the participants performed two regular breathing³ tasks: one with music played on the earbuds and one without music. We then asked our participants to perform nose breathing while standing and nose and mouth breathing while lying on a nearby couch. After the lying down tasks, the participants performed some light exercises (1 min of stretching and 6 minutes of walking) to induce the breathing rate at an elevated rate. Then we asked them to rest for one minute in a seated position (Cool Down task) and breathe naturally. Finally, they performed seven controlled breathing exercises in different cycles (5-25 breaths per minute) and types (chest and abdominal breathing). Since we focus on estimating the natural breathing at the resting position, the light exercise and controlled breathing exercises are outside the scope of this paper. However, we utilized the motion sensor data from the exercise sessions to develop the head-motion filtering algorithm.

Virtual Onboarding Session Protocol. After the lab sessions, we asked our participants for a 45-min virtual session, where they attended a video call with one researcher (due to COVID-19 safety protocol). During the video call, the participants were required to perform some activities that they can perform in their daily lives, including coughing, speaking, drinking, eating, and walking. While most coughing tasks are outside the scope of this paper, we use the drinking, eating, and speech sessions as our noise class.

In Home Protocol. At home, the participants were asked to perform the six mandatory breathing tasks two times (one in the morning and one in the afternoon) a day for seven days. Among these tasks, the participants needed to perform three tasks daily (Home

³In this work, we use regular or natural breathing to define the participant's normal breathing session without any specific instructions on how to perform the breathing. For the other tasks, we gave instructions, e.g., breathing using the nose or mouth.

tasks 1, 2, and 4 in Table 2) and three breathing tasks on alternating days (Task 3a, 3b, and 5a). In addition, the participants could perform five voluntary breathing tasks (listed as seq x). The participants were asked to perform their natural, nose, and mouth breathing in sitting, standing, and lying, similar to the lab protocol.

3.1.3 Ethical Considerations & Compensation. All participants agreed to participate in the study by signing informed consent. We also informed the participants that they would be audio-recorded whenever they wore the earbuds. The only way to stop the recordings is to keep the earbuds in the corresponding wireless charging box. The participants received \$400 for their participation.

3.2 Annotation & Ground-Truth Calculation

We conducted one of the most comprehensive data collection studies for breathing rate monitoring using earbuds. At the end of the study, we collected more than 2500 sessions (≈ 45 hours) of breathing data collection. We needed to annotate all the chestband signals for the ground truth calculation. However, the annotation from the chestband is not always reliable (due to motion or loose connection to the skin); therefore, we also performed audio data annotation for all tasks. We developed a portal for the annotation (Appendix B).

3.2.1 Chestband Data Annotation Procedure. The chest motion tracked by Zephyr produces a sinusoidal breathing signal, where valley-to-peak indicates an inhalation and peak-to-valley means an exhalation. An example of a breathing signal is depicted in Figure 15. We used a semi-automated procedure for the annotation of the breathing signal. We first used a peak-detection algorithm to identify the peaks and valleys of the breathing signal automatically. Since the automated algorithm is unreliable and can miss the peaks and valleys (Figure ??), multiple reviewers later reviewed the automatic labels. The annotators fixed the missing peaks and valleys and deleted erroneous annotations. When no consistent peaks or valleys are visible in the signal, the annotators marked those sessions as ‘Not Annotatable’, and we discarded the sessions. Most chestband waveforms were annotated by the lead researchers, who have multiple years of experience in breathing data collection using Zephyr. In addition, we hired an external annotator to annotate the tasks from the In-home sessions. We provided multiple training to the external annotator, and the lead researchers later reviewed each annotation and made modifications in case of incorrect annotation.

3.2.2 Audio Data Annotation Procedure. An external annotator annotated all audio data from all breathing sessions using a Spectrogram-based visualization in the annotation portal (Figure 16). The annotator annotated the breathing phases (e.g., inhale/exhale) and noises (e.g., background noise, speech) by listening to the audio. The external annotator received multiple training, and the lead researchers reviewed each annotation.

3.2.3 Motion Sensor Data Annotation. We also annotated motion sensor data collected in the lab sessions for developing our motion filtering algorithm (Figure 17). We annotated the sessions by visualizing the signal observed in the Gyro-X axis. Most breathing sessions from the lab sessions were marked as ‘Motionless’ segments after visualizing, as the participants were instructed not to move. However, on rare occasions, the participants had some abrupt head

motions, which we annotated as ‘Motion’ segments. We also annotated the head-motion and light activity tasks (walking, eating, drinking, and speaking) as ‘Motion’.

3.2.4 Multimodal Annotation Review. We also put extra effort into retaining the most sessions collected in the home. In case of confusing breathing waveform from Chestband, the lead researcher utilized the annotation portal’s multimodal annotation capability to visualize the chestband, audio, and motion signals together. In case of bad quality chestband signal, we marked that as bad and used an audio signal to calculate the breathing rate (Figure 18).

3.2.5 Ground Truth Calculation. From the chestband annotations, we first calculated the median breathing cycle duration (m_c) by taking the median peak-to-peak duration. From the audio annotations, we calculated the median breathing cycle duration (m_c) by taking the median inhale-to-inhale duration. In both cases, the ground truth breathing rate (BR_{gt}) is calculated as $BR_{gt} = \frac{60}{m_c}$. We mostly use the annotations from the chestband as our ground truth rate.

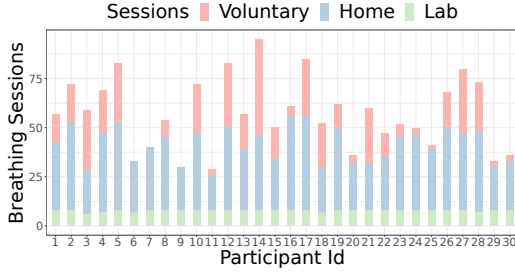
3.3 Participants & Breathing Rate Statistics

Our 30 participants (15 male and 15 female) came from diverse backgrounds and were aged between 22–58 with a BMI range of 17 to 45. Most participants were self-reported as healthy with no known significant health conditions. Most participants ($N=25$) are active earphone users and use earphones for 1–4 hours per day to listen to music, answer phone calls, watch videos, and work. In total, our participants performed 2635 regular breathing sessions. However, the annotators were able to annotate 1736 breathing sessions from the lab ($N=251$), home ($N=1003$), and voluntary ($N=482$) sessions. The participants were highly compliant with our seven-day protocol, with some actively participating in the voluntary data collection. The number of sessions each participant performs is presented in Figure 1a. On average, participants performed 70 breathing sessions during our study, with 43 (P6) being the lowest and 94 (P27) being the highest. We also have a diverse range of ground-truth breathing rates. The summary of breathing rate is depicted by the box plot in Figure 1b. Although we recruited participants with no underlying disease conditions, most had normal breathing rates (10–25 BPM). However, both P16 and P26 had abnormally low breathing rates (<10 BPM in most cases). Since detecting abnormal breathing rates is out of the scope of this paper, we only considered those sessions where the breathing rate was between the 10–30 BPM range and filtered the sessions that were outside the normal breathing range. At the end of the filtering, we use the 1531 breathing sessions in the normal breathing range for developing our models and evaluation.

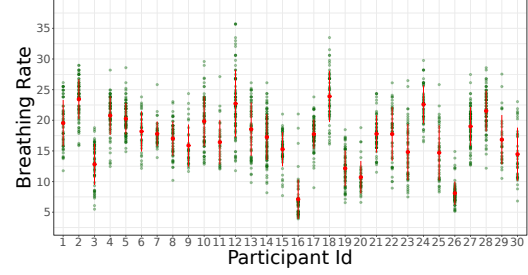
4 SYSTEM DESIGN CONSIDERATIONS

We have the following design goals for a breathing rate monitoring system based on some observations from in-home breathing sessions and the limitations of earbuds:

(1) **Multimodal System:** After observing the home data, we realized the necessity of a multimodal system. Figure 2 shows two examples from the collected data in natural environments. Although some participants tried to follow the protocol rigorously, we noticed that head motion is difficult to avoid even in the resting position.



(1a) Breathing Sessions performed by each participant



(1b) Breathing Rate Range for each Participant

Figure 1: Participant Compliance and Breathing Rate Statistics

As seen in Figure 2a, the participant was performing their natural breathing properly; however, they moved slightly between 7–10 seconds. We can not utilize such sessions by only tracking the breathing motions as it can produce inaccurate results due to the motion artifact. After inspecting the motion sensor data, we noticed that approximately 63% of the sessions had medium to high-intensity head movements that can last from 0.9 sec to 56.9 sec. Such a high amount of motion artifacts, even in the controlled breathing sessions, demonstrates the challenges of breathing rate monitoring in the natural environment. Since we need to filter out the non-breathing movements, the motion-sensing-based algorithms will not be able to retain the most valid breathing sessions due to the sudden and accidental head motions during the task. On the other hand, we also noticed that while some participants did not move significantly during the tasks, their breathing sounds were not audible or masked by noises (Figure 2b). Breathing sounds have low-frequency (20-1300 Hz) [64] sounds; so any background noises will always mask them. Additionally, some users can be silent breathers, which may not produce enough energy during breathing. From our home sessions, we identified that annotators could not annotate the audio sessions approximately half of the time due to background noises or weak breathing sounds. Audio data alone will not help to track the breathing rate on such occasions. Hence, we argue that a single modality of sensors might not be ideal for monitoring breathing rates in the natural environment; we need to utilize both audio and motion sensors to estimate the breathing rate to attain good accuracy and session retention rate.

(2) **Good Accuracy and High Retention Rate:** While aggressive filtering on the noisy data could improve the accuracy, it will discard a majority of the valid breathing session. We envision users using the system to check their respiration rate quickly. However, if the system cannot generate breathing rate estimations, even with reasonable efforts from the user, the system will not be usable. Therefore, we prioritize the session retention rate more than the accuracy itself. Our goal here is to improve the retention rate while maintaining good accuracy. Both U.S. Food and Drug Administration (FDA) [40, 66] and Consumer Technology Association (CTA) [13] recommend a maximum MAE of ± 2 BPM for any breathing rate monitoring system. Therefore, instead of aggressive noise filtering for high accuracy, our goal is to produce a breathing rate with relatively good accuracy and a higher retention rate.

(3) **Lightweight System:** We also want to develop a lightweight system to implement on the earable device without requiring the data to send the phone. Implementing the breathing tracking system

on the earbuds can reduce the networking and memory overhead. Earbuds have limited processing power and memory; therefore, we prioritize motion sensors over audio. Signal processing on motion sensors takes less computing power than audio processing. Therefore, we only run the audio module if the motion sensor fails to estimate the breathing rate from the signal.

5 MULTIMODAL BREATHING MONITORING SYSTEM

Following our design goals, Figure 3 represents our proposed approach for breathing rate tracking in an active breathing session. Since the breathing rate is measured over a minute, we assume the user will perform at least a 1-minute breathing session. Once the user initiates the session, the system records the session by using both motion and acoustic sensors. As depicted in Figure 3, we have two separate pipelines in our system. We use a lightweight motion-based breathing rate estimation in one pipeline that leverages the 6-axis IMU signals from the earbuds to estimate the breathing rate. On the other one, we use an ML-based approach that utilizes the audio recordings from the earbuds for breathing rate calculation. If the motion-based algorithm can not determine the breathing rate after processing all windows, the system then performs the audio pipeline to estimate the breathing rate. In the following section, we first discuss our motion-based pipeline (Section 5.1). Then we discuss the audio-based algorithm that uses a machine learning-based approach to calculate the breathing rate (Section 5.2).

5.1 Motion-Based Breathing Rate Estimation

As explained earlier, breathing generates a very minute head movement that the motion sensors in the earbuds can capture. Previous works [2, 62, 63, 67] showed that it is feasible to extract breathing motions from the motion generated on earbuds. We use a similar signal processing-based pipeline described in RRMonitor [2] and InstantRR [63] for the rate estimation using the earbuds. The details of the motion sensor-based pipeline are shown in Figure 3 and 4. The activation of the motion sensor-based algorithm is dependent on the head motion count (M_c). If $M_c > T_h$, then we discard the corresponding 30s window. Since we have a step size of 2s, the system tries to identify a few good windows from the 60s buffer. At the end, it estimates the motion-based breathing rate (BR_{motion}) by taking the median of all the valid measurements. The BR_{motion} is initially set to 0, and if the system does not find any good windows from the 60s buffer, then it activates the audio pipeline.

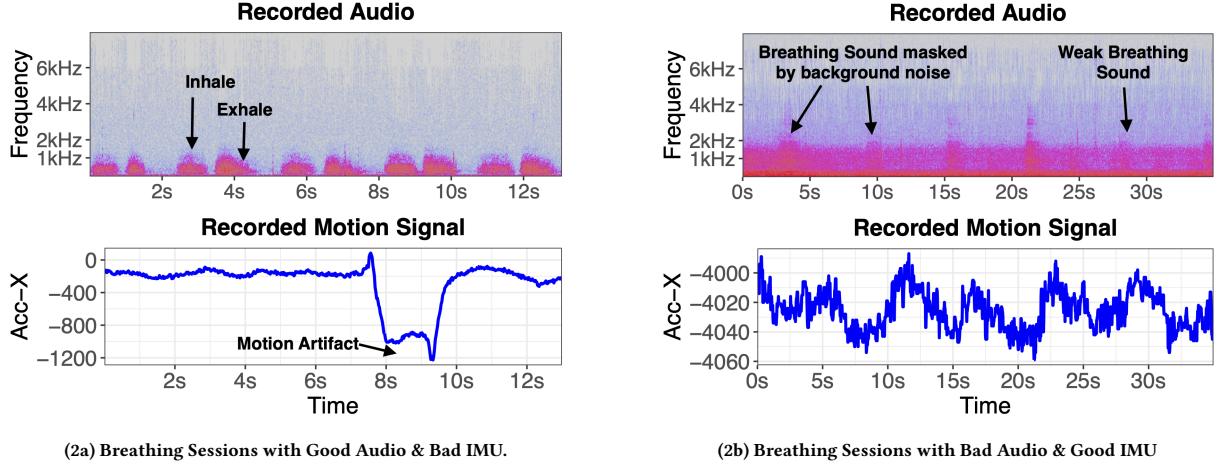


Figure 2: Audio and IMU data Quality in Uncontrolled Settings

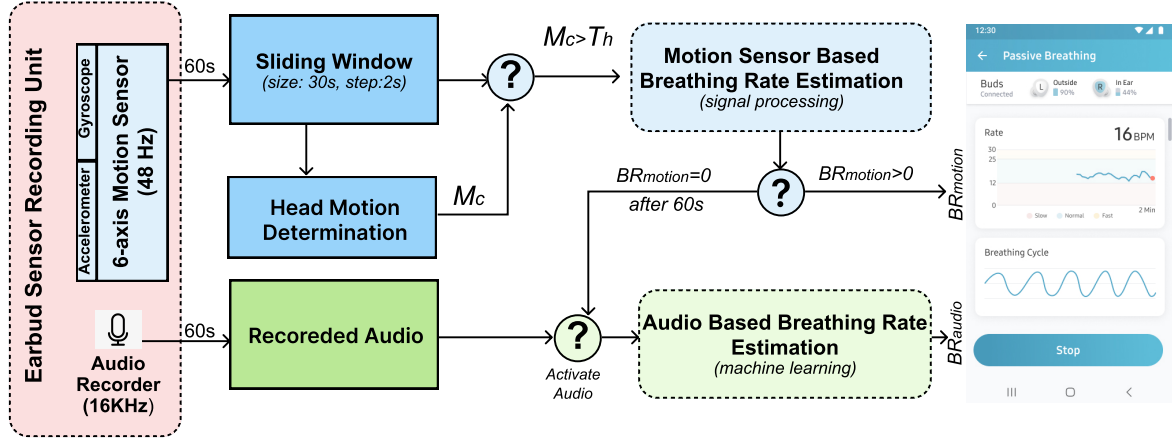


Figure 3: Our proposed breathing tracking system using earbuds. Blue components indicate the motion sensor-based pipeline, and green indicates the audio-based pipeline. The breathing rate is displayed on a mobile app.

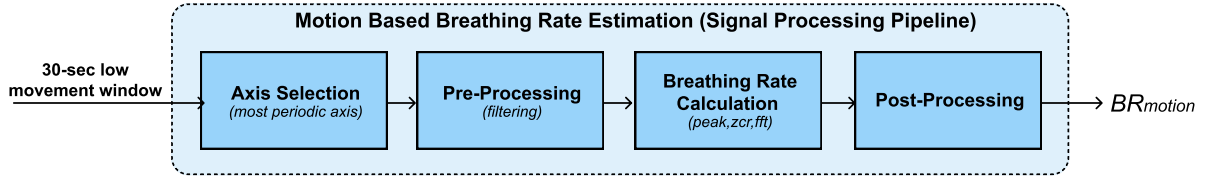


Figure 4: Detailed pipeline of motion sensor-based breathing rate estimation

5.1.1 Sliding Window & Step Size. While larger windows can generate more accurate breathing rate predictions [3, 39]; we use a 30s sliding window for estimating the breathing rate, as previous works showed that a 30s window yields reasonably good performance [2, 39]. Additionally, it enables us to have multiple predictions, so the system is robust to slight head motion. Finally, we use a step size 2s to get at least 15 predictions from the 60-sec window.

5.1.2 Head-Motion Estimation. Head motion filtering is crucial for the motion-based pipeline, as a very light non-breathing motion can contaminate the quality of the signal. Therefore, signal-processing

algorithms need to filter out the motion artifacts. To filter motion artifacts, we use a simple logistic regression classifier with only ten features to calculate the number of motions in the sliding window. Since head movements are more instantaneous, we use a small sliding window of 1s with 50% overlapping inside the 30s window. We then use the features from each 1s window to determine if the window has any motion. Finally, if the total number of 1s windows with movements detected (M_c) is greater than threshold T_h , the system discards the window and processes the next one. We utilize the breathing and non-breathing sessions to train the filtering classifier.

# Features	Logistic Regression			Random Forest-20		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score
Top-3	98.1%	92.9%	95.5%	96.8%	94.0%	95.4%
Top-5	98.1%	93.3%	95.6%	96.9%	94.5%	95.7%
Top-8	98.1%	93.4%	95.7%	97.2%	95.3%	96.2%
Top-10	98.1%	93.4%	95.7%	97.0%	95.5%	96.2%

Table 3: Performance of Logistic Regression and Random Forest classifier across different number of features.

From the motion sensor annotation (as described in Section 3.2.3), we were able to extract 34431 ‘Motion’ and 27422 ‘Motionless’ windows (1s length), which we use to train the classifier. We extract only ten statistical features from the time domain to avoid expensive calculations in the pipeline. We identify the ten features by performing the correlation-based feature selection (CFS) algorithm [26] on more than 200 most commonly used time domain features for the activity recognition [4]. The top-5 features are listed in Table 4. We experimented with Logistic Regression and Random Forest (with 20 trees) to evaluate the motion detection performance. We performed leave-one-subject-out (LOSO) cross-validation on our dataset, and the performance of the classifiers is reported in Table 3. While the Random-Forest algorithm performs slightly better, we use logistic regression to keep the signal-processing pipeline lightweight.

5.1.3 Axis Selection. We follow the similar axis selection procedure described in InstantRR [63] and BioWatch [29] to determine the *most periodic axis* from the 6-axis accelerometer and gyroscope for breathing rate estimation. We simply perform the following steps on each axis: 1) apply a median filter and then normalize the signal, 2) apply a Butterworth Bandpass filter with 0.13Hz lowcut and 0.66Hz highcut frequency, and 3) perform the Fast-Fourier transform (FFT) algorithm on the filtered signal. The axis with a maximum frequency value yields the most periodic axis with the largest variation. The rest of the part of the algorithm uses the selected axis for pre-processing and breathing rate estimation.

5.1.4 Pre-processing. We first apply a median filter to smooth the signal in the preprocessing step. We then perform normalization on the filtered signal and use a 3^{rd} order Butterworth Bandpass filter with 0.13Hz lowcut and 0.66Hz highcut frequency (corresponds to the breathing rate between 8 to 40 BPM). We then apply a 2nd order Savitzky-Golay [80] filter for further smoothing the signal. The preprocessing and filtering steps are depicted in Figure 5a–5d.

5.1.5 Breathing Rate Estimation Algorithm. As the filtered signal resembles a breathing cycle, we use the following three well-established algorithms [2, 63] to extract the respiratory rate:

- (1) *Zero-Crossing Based Algorithm (BR_{zcr}):* We can easily calculate the breathing rate by identifying the zero-crossing points in the filtered signal and taking the time between two zero-crossing points. We take the median time of the zero-crossing duration to estimate the breathing rate: $BR_{zcr} = 60 / \text{median}(\text{zero-crossing-duration})$.
- (2) *FFT-Based Algorithm (BR_{fft}):* We can apply the FFT algorithm to the filtered signal to compute the frequency domain representation. The breathing rate can be calculated by selecting the frequency of the highest amplitude (peak) within the 0.13 Hz and 0.66 Hz frequency range (7.8 to 40 BPM). In this approach, the rate is calculated as $BR_{fft} = 60 \times \arg \max FFT(x)$.

Top Feature Names
derivative_acc_mag_mean
derivative_acc_mag_sma
derivative_acc_mag_max
derivative_accZ_sma
acc_mag_sma

Table 4: Top-5 features for motion detection (sma: signal magnitude area, mag: magnitude)

(3) *Peak Detection Based Algorithm (BR_{peak}):* We use a peak detection algorithm to find the peaks and valleys in the filtered signal. Similar to the ground-truth breathing cycles, each valley-to-peak indicates an inhale cycle, and the peak-to-valley indicates an exhale cycle. In an ideal breathing cycle, peaks and valleys must occur in alternating order. However, the algorithm can detect false peaks or valleys due to the small noises in the derived breathing cycle. For that reason, we remove the false peaks and valleys if there are multiple peaks in between two valleys or multiple valleys in between two peaks. When there are multiple peaks in between valleys, we select the ones that are closest to the nearest valley. Similar to the ground-truth breathing rate calculation, we can estimate the rate by taking the median peak-to-peak distances. In this approach, the rate is calculated as $BR_{peak} = 60 / \text{median}(\text{peak-to-peak-duration})$.

5.1.6 Post-Processing. Even after head-motion and noise filtering, the breathing rate estimation can be incorrect due to the lack of breathing motions. In some instances, the breathing motions can be so small that none of the algorithms can accurately predict the breathing rate from the motion signals. In other instances, the head-motion filtering algorithm might fail to detect a small head motion, generating an incorrect prediction. In the post-processing, we filter some noisy predictions by taking the standard deviation of three algorithms. Ideally, the three breathing rate algorithms should predict a similar number. However, if the predictions have a higher standard deviation, that may mean the performance of the algorithms is inconsistent for that particular window. On such occasions, we discard the corresponding prediction. If the estimation quality (e_q) = $\text{std.dev}(BR_{peak}, BR_{zcr}, BR_{fft}) > 3$, we discard the corresponding window. If we set e_q lower, then the estimation will have better accuracy, but it will discard more windows. If we set e_q higher, then the estimation will have more error, but it will retain more windows. After some experimentation, we set $e_q = 3$.

Since we are running the motion-based breathing rate pipeline every 2s, so we will have 15 window level estimations in the 60-sec window. In the end, we take the median of the unfiltered predictions as BR_{motion} . If the system could not find at least two valid predictions, it discards the output from the 60s window and sets BR_{motion} as 0. On such occasions, an audio pipeline will be activated.

5.2 Audio-Based Breathing Rate Estimation

Breathing audio has a unique pattern than any other low-frequency sound, and several previous works showed that it is possible to classify breathing sounds from other non-breathing sound [31, 37, 64]. Figure 6 presents our ML algorithm pipeline that utilizes the earbuds’ audio sensors. Compared to previous work, we use a transition-based approach to calculate the breathing rate, which is lightweight enough to fit in a low-resource device like earbuds. In our approach, we use a machine learning classifier to detect the

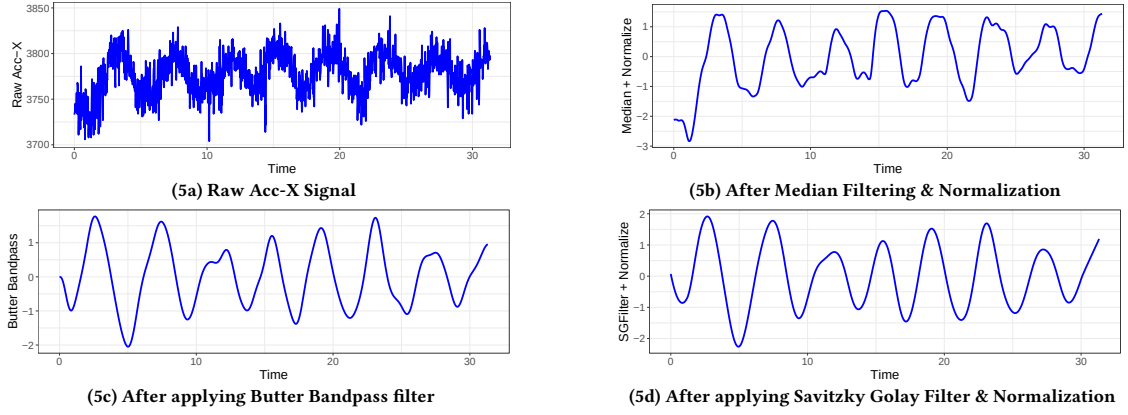


Figure 5: Steps of Noise Filtering and Signal Pre-processing

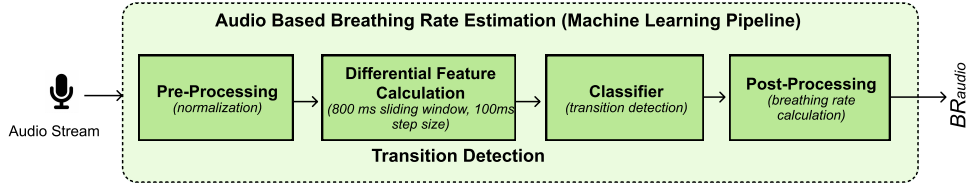


Figure 6: Detailed pipeline of audio-based breathing rate estimation.

start of the breathing events and use that for breathing rate calculation. Figure 7 shows the audio spectrogram of breathing sessions with and without background noises. As seen in Figure 7a, there is a small transition window from one breathing phase to another breathing phase (around 0s, 4s, 6s, and 10s). We use the term ‘*Transition*’ to define the change of breathing phases, from inhale to exhale or exhale to inhale. We only need to identify the start of the breathing phases or the transitions for breathing rate calculation. As shown in Figure 7a, the transition phase always indicates a change of energy. Most of the time, the start of the breathing phase has higher energy than the end of the previous breathing phase. In our pipeline, we use an ML-based approach for this transition detection. The details of the pipeline are explained below:

5.2.1 Windowing and Step Size. Since transitions can be instantaneous, we use an 800ms with a 100ms step size to identify the transitions in the breathing signals. We divide the 800ms window into two 400 ms windows, which we call *left* and *right* windows. We then calculate the difference between the two 400ms windows and call it *differential features*. Finally, we use these differential features to detect the transitions and calculate the breathing rate. The benefit of using the differential features is it negates small windows with similar patterns and only recognizes sharp changes in sound energy. Therefore, we do not need additional noise filtering.

5.2.2 Feature Description. For audio-based features, we experimented with 51 time, frequency, and spectral density features previously used for wheeze detection [10]. The list of features is listed in Table 5 and 6. From the audio data, we observed low background noise could have transitions similar to breathing (Figure 7b). However, the average band power in different lower frequency ranges could be unique to breathing signals. Therefore, we use absolute

(average band power) and relative density (relative band power compared to the total power of the signal) in six different frequency ranges. The frequency ranges are listed in Table 6.

5.2.3 Dataset Preparation. To train the classifier, we use the audio data from slow-breathing sessions (Task 4 in Home session Table 2). We observed that the breathing audios from slow breathing sessions are similar to the natural breathing in the home environment with similar environmental variations. However, the participants used face masks during the lab sessions, attenuating the breathing sounds so lab data could underestimate the model’s performance with home data. We could not use the data from the home sessions as it will overestimate the performance of the breathing rate in the home data. Therefore, slow-breathing tasks are ideal for classifier training. In the slow breathing task, participants performed breathing at a 5 BPM rate. We provided animation to assist the participants with the 5BPM breathing rate. During the audio annotations, we annotated the audio and marked the start and end of the breathing phases. We then take an 800ms window around the beginning of the breathing phase and consider them as *transition* class. The rest of the cycle is considered as *breathing*. We use the audio from virtual sessions (speech, eating, and drinking tasks) for the *noise* class. In the end, we have 1964 transitions, 1642 breathing, and 75 noise samples. After taking 800ms sliding windows with a 100ms step size, we have 5753 breathing, 5851 transition, and 3140 noise segments.

5.2.4 Classifier & Performance Evaluation. Since earbuds have limited memory, the size of the ML model is crucial for our pipeline. Therefore, we experimented with several lightweight models, including random forest classifiers with 10, 20, and 50 estimators, with a maximum depth of 10. We also included a multilayer perceptron (MLP) model of $(15 \times 30 \times 15)$ network size. We use Adam

Domain	Features
Time	Mean, Median, Standard Deviation, Skewness, Kurtosis, SPLLevel, Quartile Range
Frequency	Spectral Centroid, Spectral Spread, Spectral Rolloff, Spectral Flatness, Spectral Skewness, Spectral Kurtosis, Spectral Std.Dev, Chroma (12), MFCC(13)

Table 5: Time & Frequency Domain Feature List

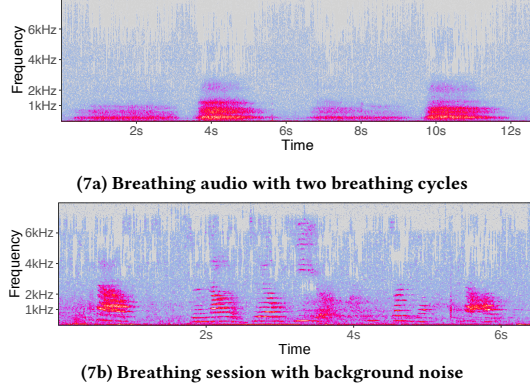


Figure 7: Spectrogram of breathing sessions with or without noise

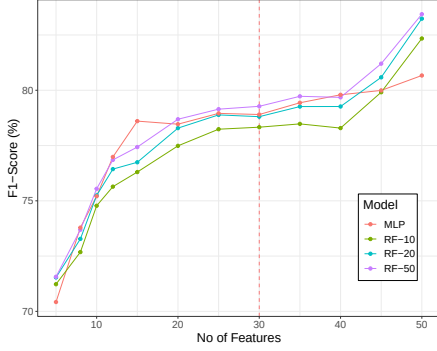


Figure 8: Performance of Transition Detection

optimization solver [36] and logistic function as the activation function for the MLP classifier. To reduce the complexity of the models, we use the Correlation-Based Feature Selection (CFS) algorithm for feature selection. For evaluation, we use a leave-one-subject-out cross-validation, where we put the data from a single subject into the test set and the rest into the training. The performance of the transition detection classifier is presented in Figure 8. As seen in Figure 8, more features can improve the detection accuracy. However, calculating 50 features within 100ms is not feasible on buds. After experimentation with earbuds implementation, we select the top 30 features in the final implementation. We use the MLP model as it gives the smallest model size ($\approx 45\text{KB}$).

5.2.5 Post-Processing & Breathing Rate Calculation. Since we are generating a prediction every 100ms, there will be small clusters of different classes, and in an ideal case, we will have multiple clusters of transitions in the 60s audio recordings. We use a clustering approach to identify the start and end of the clusters. The difference between the adjacent clusters' start times is defined as the 'transition' duration (d_t). The breathing rate is calculated from the median transition duration, as $BR_{audio} = \frac{60}{median(d_t)}$.

Type	Frequency Range
Absolute Density	[100-300], [301-500], [501-800], [801,1000], [1001-2000], [2001-5000]
Relative Density	[100-300], [301-500], [501-800], [801,1000], [1001-2000], [2001-5000]

Table 6: Spectral Density Features

We use several estimators to determine the audio quality and breathing rate prediction. In a normal breathing range (10–30 BPM), the 60s audio should be at least 20 cycles. The no of the clusters (N_c) is a good indicator of breathing cycles, which we can use for quality estimation. If N_c is higher, it will filter more sessions and the accuracy will be better. Since we have a 100ms sliding window, we can expect multiple transitions detected in a cluster; therefore, we also use the size of the cluster (S_c) as another quality parameter. If S_c is small, then it could indicate that the classifier might have detected a false transition; therefore, the accuracy will be lower. We also use another quality parameter called noise to breathe ratio (NBR), which we define as:

$$NBR = \frac{\text{No. of Noise Window Detected}}{\text{No of Breathing Window Detected}}$$

Since noise signals can mask the breathing audio, high NBR could mean that the breathing audio is too noisy for breathing rate prediction. If we keep NBR smaller, it will be more sensitive towards noises, and the accuracy will be better. While the quality parameters could be very intuitive, we identify the thresholds for N_c , S_c , and NBR after performing some experiments. We determined that 1) $S_c \geq 3$, which means the transition detector should detect at least three transitions to predict the breathing rate, 2) $N_c > \geq 10$, since we are targeting to predict 10–25 BPM range the detector should detect at least five cycles or ten clusters for accurate estimation, and 3) $NBR < 2$, which is the no. of noise counts should not be more than twice than the breathing segments. Such combinations of quality parameters give us the best balance between the accuracy and retention rate. Section 6.3 presents some experimentation with few combinations of these parameters. If the 60s recording does not meet quality, the ML- pipeline predicts BR_{audio} as 0.

6 EVALUATION

This section discusses the evaluation of our motion and audio-based algorithm in 10–30 BPM breathing rate range.

6.1 Metrics

We use the following two metrics for evaluating the performance:

- (1) **Mean Absolute Error (MAE):** Mean Absolute Error (MAE) is the most commonly used performance metric reported in previous works [37, 39, 66, 67]. Additionally, the Consumer Technology Association (CTA) guideline recommends using MAE as the performance evaluation criterion [13]. The mean absolute error is defined as the average of mean absolute differences between BR_{gt} and BR_{est} :

$$MAE = \frac{\sum_{i=1}^n |BR_{gt_i} - BR_{est_i}|}{n}$$

Here BR_{est} is the estimated breathing rate, which could be either BR_{audio} or BR_{motion} .

- (2) **Retention Rate (RR)**: The session retention rate is defined as the percentage of sessions the system was able to retain. If we have n total sessions and the system was able to predict the breathing rate in n_p sessions, then the retention rate is defined as $RR = \frac{n_p}{n} \times 100$.

While evaluating the parameters and algorithm, we emphasize balancing accuracy and retention. Although we can achieve excellent MAE by filtering more sessions, we focus on attaining reasonable MAE by retaining more sessions. Since the FDA [40] and CTA [13] recommend an accuracy of ± 2 BPM, we tried to achieve < 2 BPM MAE in most sessions.

6.2 Key Parameters of Motion-Based Algorithm

We have several key parameters in our motion sensor-based algorithm. In this section, we present the performance of our motion sensor-based algorithm with different parameters. Since our goal is to achieve better performance in uncontrolled settings, we use a subset of the in-home data for evaluating the key parameters in the motion-based pipeline. We only use the four mandatory tasks from the in-home sessions (Tasks 1, 2, 3a, 3b, and 4 in Table 2).

6.2.1 Head Motion Filtering. Motion count (M_c) is a crucial parameter for our algorithm; we filter bad windows based on the number of head motions detected. Since the head motion prediction model is trained on the lab data, we use the data from home sessions to evaluate the head motion filtering model and set up the motion count threshold T_h . In this analysis, we use the window level (in every 2s) predictions for identifying the threshold. The performance of the motion-based breathing rate estimations is presented in Figure 9. As expected, the performance improves with a stringent threshold. If $M_c < 1$, then the system can estimate the breathing rate with MAE < 2 BPM for most sessions. However, most windows will be discarded, and the average retention rate is 33.4%. Since our goal is to give a more accurate performance with some room for a lower retention rate, we select T_h as 2. We can cover some of the retention rates by using the audio-based pipeline. However, we notice that the parameters can be adjusted based on activity. If the user is in a lying position, we can adjust the threshold to up to 4 and can achieve a 60% retention rate with a slightly degrading performance (MAE = 2.02 BPM).

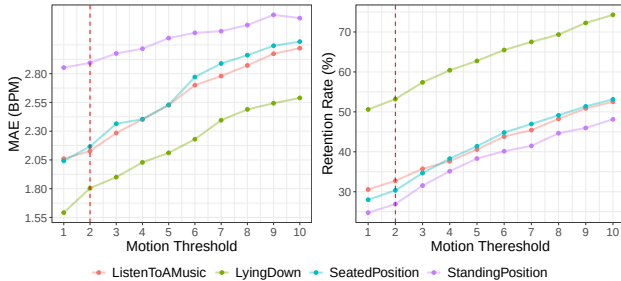


Figure 9: Performance with Head Motion Filtering Across Different Home Tasks and Motion Thresholds. We use BR_{zcr} as the estimated breathing rate. A small threshold improves the accuracy while discarding more sessions. Motion threshold (T_h) gives the best combination, retaining most windows and keeping the MAE < 2 BPM.

Task	MAE	Retention Rate
Music	3.63	66.5%
Lying	2.83	73.3%
Seated	3.53	63.2%
Standing	3.40	54.8%

Table 7: Performance of Acceleration Based Head Motion

Since the head motion prediction model is trained on the lab data, we use the data from home sessions to evaluate the motion filtering model. We compare the performance of the logistic regression model with a magnitude-based algorithm used in previous works [67]. In the magnitude-based approach, the algorithm determines the magnitude of the accelerometer values and discards a window if more than 3-5% values have a magnitude greater than 10 m/s^2 . Since accelerometer also contain the gravity component, the motionless accelerometer values should be closer to 9.8 m/s^2 . We use the same approach in our in-home sessions and observe that accuracy is worse than the Logistic Regression based approach. The performance of the magnitude-based algorithm is shown in Table 7. Although the approach retains more data, the MAE is greater than 3 BPM, which is beyond the acceptable accuracy.

6.2.2 Best Algorithm. Although we use all three algorithms to estimate the quality of the prediction, we need to select one algorithm to show it to the user. Figure 10 shows the performance of three breathing rate algorithms. Based on our evaluation, we identified that BR_{zcr} performs more consistently across different breathing rates. BR_{zcr} is also the most lightweight algorithm [2], so it seems more suitable for the standalone algorithm in the low-resource earbuds. Considering the performance, we select BR_{zcr} as the output of the motion-based pipeline (BR_{motion}).

6.3 Balancing Accuracy and Retention Rate

To identify a good balance between the accuracy and session retention rate, we experimented with regular breathing sessions in the home environment in sitting, lying, and standing positions. We have two quality parameters from the motion-based pipeline (Motion Count Threshold T_h and estimation quality e_q) that filter the noises of the motion signal and improve the accuracy. We have three quality parameters from the audio-based pipeline: 1) no of clusters (N_c), 2) minimum size of clusters (S_c), and 3) Noise-to-Breathing Rate (NBR) to handle the noise data. We selected five settings ranging from *very aggressive* to *very minimal* filtering of noisy data. We use different combinations of parameters from our multimodal pipeline for these five settings. The performance of different combinations is presented in Table 8, and the MAE distribution is presented in Figure 11. As can be seen in Table 8 and Figure 11, a ‘*very aggressive*’ noise filtering reduces the session retention rate (combined RR: 40.6%), while improving the MAE (combined MAE: 1.36, Std.Dev: 1.91). On the other hand, ‘*very minimal*’ filtering retains most sessions (combined RR: 96.81%), which reduces the MAE (combined MAE: 2.88, Std.Dev: 3.08) as the signal can have noise. We can get the best balance between the MAE and session retention rate with the ‘*moderate*’ filtering settings. While it keeps the MAE (combined MAE: 1.93, Std.Dev: 2.37) lower than the FDA-approved limit, it retains a reasonable number of breathing sessions ($\approx 75\%$).

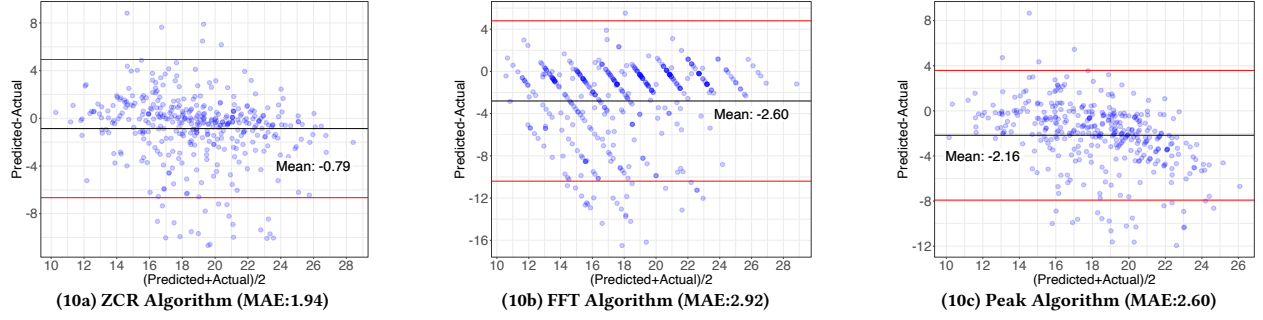


Figure 10: Bland-Altman Plot showing the agreement between the median ground truth and respiratory rate estimation algorithm. Each point represents a single session. ZCR algorithm seems to be the most consistent.

Filtering	Thresholds					Performance					
	Motion		Audio			Motion		Audio		Multimodal	
	T_h	e_q	S_c	N_c	NBR	MAE	RR	MAE	RR	MAE	RR
Very Aggressive	1	1	4	15	1	1.46	28.90%	1.08	19.16%	1.36	40.60%
Aggressive	1	2	4	10	1.5	1.64	37.80%	1.54	32.77%	1.53	57.14%
Moderate	2	3	3	10	2	1.96	42.80%	1.7	56.40%	1.93	74.60%
Minimal	5	4	3	5	3	2.44	55.10%	2.11	71.40%	2.35	87.70%
Very Minimal	10	5	2	3	5	2.86	67.70%	2.7	87.90%	2.88	96.81%

Table 8: The Performance of the multimodal pipeline with different noise filtering settings. MAE presents the BPM error.

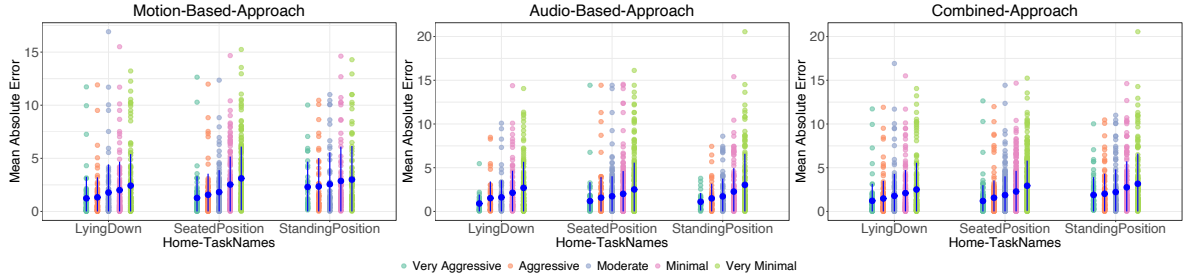


Figure 11: Mean Absolute Error distribution for different noise filtering settings on the regular breathing data collected in the home.

6.4 Performance of Breathing Rate Estimation

We present the final performance of the audio-based, motion-based, and multimodal pipeline in this section:

6.4.1 Overall Performance Comparison. The MAE and retention rate of the motion, audio, and multimodal pipeline are presented in Table 9. The Bland-Altman plot for all pipelines are presented in Figure 12, which also represents the prediction quality. As seen in the table, both algorithms perform reasonably on both recordings from both controlled and uncontrolled environments. In all sessions, the retention rate increases by using the multimodal approach. While using only one type of sensor, we can retain 20-70% of the sessions, combining both approaches can achieve a 50-88.6% retention rate in most sessions. The combination of both sensors can retain 75.1% sessions with an MAE of 1.87 (Std.Dev 2.34). As Figure 12 suggests that the approaches performs well for the full 10–30 BPM range. While sometimes we notice that the prediction has a higher error (>10 BPM), most cases have less than 2 BPM errors. We also added the performance with two baseline estimators: one that predicts

the *average* of the breathing rate of the participants (18.28), and another one that predicts a *random* breathing rate in 10–30 BPM range. As can be seen in Table 9, all our pipelines perform better than both average (MAE: 3.50, Std.Dev: 2.43) and random predictor (MAE: 6.12, Std.Dev: 4.17). We performed a statistical t-test on the average estimator with the performance of the multimodal approach, and the difference is statistically significant ($t = -16.96$, p -value < 0.001). Currently, the multimodal approach favors the motion-based pipeline, therefore, the MAE is expected to be closer to the motion-based MAE. However, the multimodal MAE could improve if the audio-based pipeline has better accuracy, and the MAE could degrade if the audio pipeline has lower accuracy. On two tasks (underlined in Table 9), we noticed that the multimodal approach performed worse than the audio and motion-based approach because the audio model introduced some noisy data. We have one audio sample with a high error (>10 BPM) in both cases, which reduces MAE. Overall, the multimodal approach seems very promising as it improves session retention rate in all of the breathing sessions, while keeping the accuracy < 2 BPM.

Settings	Task Information			Baseline (MAE)		Motion Based		Audio Based		Multimodal	
	Postures	TaskName	N	Average	Random	MAE	RR	MAE	RR	MAE	RR
Lab*	Sitting	Nose Breathing	25	4.24 (2.59)	7.33 (4.47)	2.02 (2.43)	52%	1.95 (1.91)	64.0%	1.90 (2.05)	80.0%
		Mouth Breathing	24	4.62 (2.82)	6.99 (4.76)	1.37 (2.03)	55.5%	1.60 (2.67)	62.5%	<u>1.61</u> (2.72)	83.3%
		Music	26	3.46 (2.22)	4.64 (3.81)	2.98 (3.24)	53.8%	-	0%	2.98 (3.24)	53.8%
		Regular Breathing	27	3.62 (2.36)	7.77 (4.85)	2.01 (2.84)	55.5%	1.98 (1.85)	59.2%	1.80 (2.49)	77.8%
		Cool Down	26	3.02 (1.82)	6.16 (4.51)	2.05 (2.59)	57.7%	0.78 (0.83)	46.1%	1.84 (2.33)	73.1%
	Standing	Nose Breathing	26	3.37 (2.35)	6.86 (4.58)	2.32 (2.63)	57.7%	0.94 (0.89)	53.8%	1.97 (2.36)	76.9%
	Lying	Nose Breathing	22	3.62 (2.35)	6.99 (4.37)	1.95 (2.57)	63.6%	0.92 (1.40)	63.7%	1.63 (2.30)	86.4%
		Mouth Breathing	25	3.61 (2.18)	6.08 (4.07)	1.10 (2.22)	60%	2.61 (3.28)	56.0%	1.62 (2.92)	84.0%
	Home	Regular Breathing	286	3.50 (2.44)	6.09 (4.23)	1.82 (2.07)	37.1%	1.73 (2.29)	55.9%	<u>1.87</u> (2.14)	69.6%
		Music	299	3.32 (2.37)	5.98 (4.08)	1.79 (2.18)	40.1%	-	0%	1.79 (2.18)	40.1%
	Sitting	Nose Breathing	99	3.52 (2.50)	6.89 (4.54)	2.03 (3.20)	43.4%	1.47 (1.84)	50.5%	1.74 (2.56)	73.7%
		Mouth Breathing	79	3.87 (2.49)	5.84 (4.09)	1.43 (1.94)	50.6%	2.12 (3.30)	41.5%	1.84 (2.70)	75.9%
	Standing	Regular breathing	155	3.31 (2.50)	6.32 (3.91)	2.55 (2.99)	21.2%	1.72 (2.03)	58.1%	2.21 (2.59)	75.5%
		Nose Breathing	82	3.76 (2.44)	5.00 (4.07)	3.06 (2.77)	41.5%	2.03 (2.69)	56.1%	2.21 (2.39)	71.9%
	Lying	Regular breathing	154	3.41 (2.42)	6.30 (3.99)	1.78 (2.65)	60.4%	1.61 (1.95)	55.8%	1.77 (2.50)	83.1%
		Nose Breathing	94	3.47 (2.39)	5.85 (4.26)	1.56 (2.03)	69.5%	1.71 (1.90)	40.4%	1.55 (1.90)	81.7%
		Mouth Breathing	82	3.67 (2.49)	5.81 (3.88)	1.25 (1.42)	78.4%	2.59(3.22)	32.9%	1.45 (1.74)	88.6%

Table 9: Performance of the combined approach pipeline across different tasks in a controlled and uncontrolled environment. The motion-based results from Lab sessions were filtered using a magnitude-based head motion algorithm. MAE presents the error in BPM. The standard deviation is reported in parentheses.

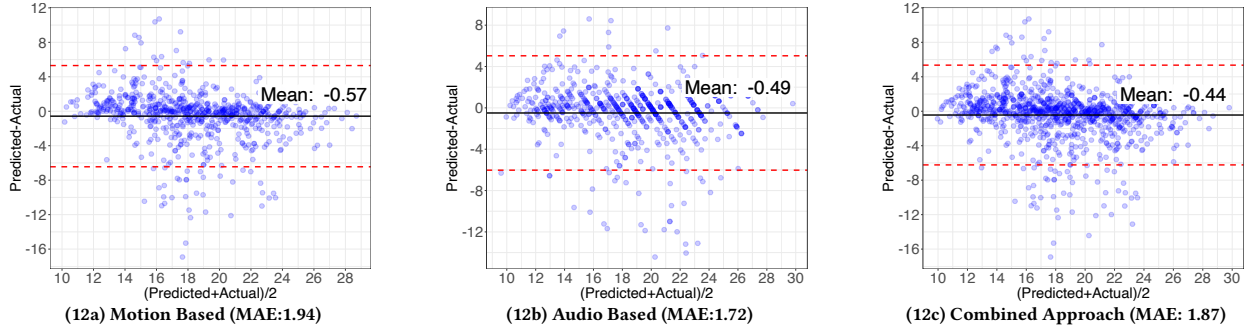


Figure 12: Bland-Altman Plot showing the agreement between the median ground truth and different pipelines in estimating the respiratory rate. Each point represents a single session.

6.4.2 Motion vs Audio. The result also suggests that the audio-based algorithm performs relatively better (MAE: 1.72, Std. Dev: 2.40, retention rate 53.1%) than the motion-based one (MAE: 1.94, Std. Dev: 2.27, at 49.4% retention rate). However, the performance difference in the performance of audio and motion-based pipelines is not statistically significant (p -value=0.42, $N=312$, ran t-test on samples where both audio and motion has a valid prediction). However, both audio (t-test, $t = -12.681$, p -value < 0.001) and motion (t-test, $t = -14.736$, p -value < 0.001) based approach performed significantly better than average estimator.

6.4.3 Lab vs Home. Since we use the lab data to train the logistic regression classifier, we utilize the magnitude-based head motion filtering algorithm proposed in previous works [2, 67]. We use a different approach to show that even in controlled breathing sessions, the existing algorithms can not retain the majority of the sessions and how the multimodal approach can be useful in improving the retention rate in controlled breathing sessions. The motion-based

algorithm performed consistently in both controlled and uncontrolled scenarios, while the audio-based algorithm performed better in the controlled environment. The performance is expected as the lab sessions were highly controlled without the presence of background noise. However, even in controlled scenarios, none of the unimodal sensors could achieve more than 70% retention rate.

6.4.4 Posture. The experimentation can be very helpful in designing the user interaction for breathing rate monitoring. As expected, the participants tend to move significantly in a standing position, and the motion-based algorithm has relatively lower accuracy and retention rate in a standing position (MAE: 2.64 at 40.1% retention rate). On the other hand, participants were more stable in lying down position, and the motion algorithm was more accurate with a good retention rate (MAE: 1.53 at 66.4% retention rate). However, audio performed consistently across different postures.

6.4.5 Nose vs. Mouth. We expected mouth breathing might produce the breathing sound with higher intensity, so the performance

will be better for mouth breathing. Surprisingly, the motion-based algorithm performed better for the mouth-breathing tasks than the audio-based algorithm. By carefully reviewing the sessions, we identified that mouth breathing signals tend to have better breathing movement, which yields better performance.

6.4.6 Music. During the music task, since we were recording the audio during the music task, the sound was recaptured by the earbud. The music audio was leaked and recorded in our recordings. The music completely masked the breathing sounds, and we were not able to remove the sound from the session. We used the voice headset profile (HSP) protocol for audio recording during the breathing sessions, and due to this protocol, the music was leaked. We could not utilize the audio recordings from music sessions. However, it did not affect the quality of the motion sensors, and the algorithm worked well with good MAE. However, in our on-device implementation, we noticed that this audio leaking did not occur. Therefore, we expect the audio algorithm to perform similarly to the other seated breathing session.

6.4.7 Performance across Participants. The multimodal approach worked differently for different participants. Figure 13 shows the MAE and retention rate for 30 participants. For example, P18 (MAE: 3.45) and P30 (MAE: 3.81) have a higher MAE with more than 50% retention rate. On the other hand, P4 (MAE: 1.16), P8 (MAE: 1.08), and P10 (MAE: 1.14) have a good MAE with more than 70% retention rate. For all but two participants, the multimodal approach was able to retain more than 50% of sessions. P20 and P26 have relatively lower breathing rates (Figure 1b); therefore, the number of samples in the 10–30 BPM range was very small. Their breathing pattern also seems slightly lighter than the rest of the subjects. This result indicates that a personalized approach might work better for the general population. We need to tune the parameters of the pipeline to handle these individual cases.

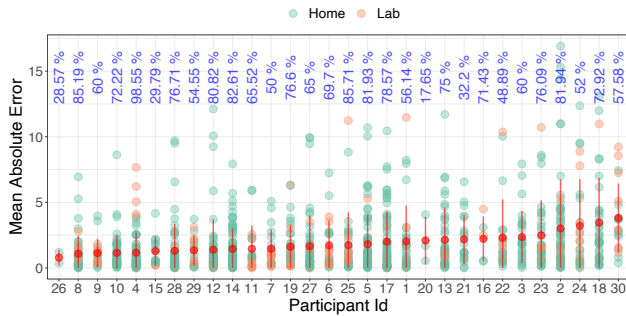


Figure 13: Participant's MAE and Retention Rate sorted in the ascending order of MAE (based on multimodal approach). The blue text on the top shows the retention rate.

7 SYSTEM IMPLEMENTATION

To determine the feasibility of the on-device implementation, we fully implemented the signal processing and machine learning pipeline on a commercially available Samsung earbud (Galaxy Buds 2). We implement all the major components of the signal processing algorithm in C to run the algorithm on the microcontroller. The

transition detection classifier is developed in Python and later converted to C using 'eml-learn' [56]. We also develop an android app to show the breathing rate continuously. The app's user interface is presented in Figure 3. The signal processing part takes 60 KB of memory, and the audio pipeline takes approximately 300 KB of memory. We also run the algorithms multiple times on two devices. The average battery consumption rate by the audio-based algorithm is 18%/hour, and the motion-based algorithm is 4-6%/hour. As expected, the audio-based algorithm consumes more memory and processing; therefore, it consumes more battery life. If no algorithm is running on buds, the battery decays by 3-5% hour. This indicates that the signal processing pipeline uses very minimal resources and can run the algorithm continuously on the earbud.

8 DISCUSSION

In this paper, we present a multimodal breathing rate monitoring system using the motion and audio sensors of the earbuds. To develop and evaluate our pipeline, we conducted a comprehensive data collection study in both *Lab* and *Home* settings. Our multimodal approach could be useful in many ways and could provide a new research direction toward continuous breathing rate monitoring:

8.1 Designing Next Generation Health Applications for Earables

Choudhury [11] argued that earables have the potential for major disruptions. In fact, the demand for earable devices is continuously increasing [7, 8, 11], and earable devices are being considered the next-generation health wearable [30, 33, 58]. As a result of this interest, there have some growing interest among the human computer interaction researchers. Several recent works explored new interaction techniques using earables including on-face interaction [86], breathing exercise [61, 83], authentication [19, 81], motion sensing [19], facial expression [20], cough sensing [38, 52, 82, 88], and other health monitoring [9, 32, 37, 43, 62]. As a next step towards this movement, we demonstrate the capability of using earbuds for breathing rate monitoring. While there exists a large number of growing works that applies a contactless approach for breathing monitoring by utilizing wireless signals [1, 16, 23, 65, 73], sonars [15, 47], and high-fidelity cameras [24, 35, 59, 69, 78] for breathing rate monitoring, earable devices can provide a more convenient and accessible solution. Our work is the first step towards designing a complete end-to-end solution, however, this work could influence the mobile computing and human computer interaction researchers to consider more applications and interaction techniques for earbuds.

8.2 Spot-Checking Breathing Rate in Different Scenarios

Breathing rate is a critical vital sign, and users may wish to check their breathing rate daily for different purposes. For example, clinicians may need to assess the patient's respiratory rate during a telehealth visit. In addition, patients with respiratory diseases often need to check their breathing rate to manage and quantify their lung condition. Several clinical studies [55] identified that the breathing rate increases before lung condition exacerbation, and tracking breathing rate could help the patients to intervene early with their

disease by taking medications. Respiratory rate monitoring at home could also help all motivated users for ‘Quantified-Self’ [42]. In such scenarios, users can quickly use their earbuds to estimate their respiration rate. While some compliance from the user is needed (e.g., sitting or lying in a quiet room with minimal head movement), our results indicate that this multimodal system can estimate the breathing rate with good accuracy and retention rate. Currently, users use manual counting or expensive devices to measure their breathing rates. Although we need some compliance from the user, the capability of using earbuds could reduce some user burdens.

8.3 User Interaction Design for Quick Respiratory Rate Assessment

The findings of this work could be used to design a better interaction for the users. Our result suggests that the pipeline has a higher error when the user is standing; therefore, standing might not be the best position for breathing monitoring. On the other hand, our result also suggests that the system can track breathing rate more accurately and with a higher retention rate in a lying down position. Therefore, the interface could instruct the user to move to a lying position for breathing monitoring. Similarly, the system might work better with no music playing on the earbuds. These results could help the designers to design a more user-friendly interface with proper instructions for the user. Additionally, our results suggest that the algorithms can not estimate the breathing rate in the presence of noise. Therefore, the designers and researchers could utilize these results and design interesting interaction techniques to guide the user to make a successful attempt. For example, when the system cannot estimate a breathing rate due to noise, the system could prompt the user to move to a different location.

8.4 Performance Comparison with Related Works

While several existing works [2, 37, 62, 67] explored earbuds to estimate the regular breathing rate, we evaluated our approach with the most comprehensive dataset covering several key variables (e.g., with or without music, nose vs. mouth). Most recent works focus more on the accuracy of the estimation, only Rahman et al. [62] reported session retention rate. In this work, while we could achieve state-of-the-art performance with aggressive filtering (MAE: 1.36), we focus more on the session retention rate to improve the system’s usability. Retention rate is a key variable for the usability of a breathing monitoring system; we hope that our work could provide useful guidelines for evaluation.

8.5 Towards Passive Breathing Rate Monitoring

An ideal breathing monitoring should be passive and should not interfere with the user’s regular activity. While our approach requires user engagement; it could be extended as continuous passive monitoring. Since the motion sensor algorithm is very lightweight and does not consume significant resources, the signal processing pipeline can continuously monitor the breathing rate without requiring interaction from the user. However, as we have seen in this study, the retention rate could be lower. Continuous audio sensing for improving the retention rate is not feasible on earbuds as it requires heavy resources and raises privacy concerns. For continuous

monitoring, we can consider using a duty-cycling approach, where the system can utilize a motion sensor the majority of the time. If it is not able to capture any valid measurements a few times per hour, it can quickly switch back to audio and record the breathing rate. However, we have not explored the algorithm’s performance in longer sessions, and we need more exploration.

8.6 Context Based Breathing Rate Monitoring

We can take a context-based approach for breathing rate monitoring. From our results, we can see that the motion-based algorithm performs better in lying-down positions. Our breathing rate in resting position are mostly stable and does not change too much during the day. An activity detection module can be placed to determine ideal condition, and active the monitoring schedule accordingly. Another useful context can be calculating the breathing rate during exercise, as it can be a helpful fitness metric. During exercise, our breathing becomes heavier, which could be picked by the audio sensor for audio-based breathing rate monitoring. Therefore, the system can determine various contexts and activate different component of our approach. For example, when the user is in a resting position, the likelihood of motion artifact would be lower, so the system can use the motion-based pipeline. When it is likely to have more head motions, the system can take short snippets of audio recordings and calculate the user’s breathing rate.

8.7 Limitations

We have a few limitations in the study. The study was conducted with participants who regularly participate in the data collection study (they are recruited via a research organization). Therefore, the participants were more compliant. In real settings, people could be much less compliant, and the likelihood of the head motions could be higher. Additionally, we only evaluated our algorithm within the normal breathing range. The utility of breathing rate monitoring will be much more useful for those who have respiratory conditions. Their breathing rate could be significantly higher or lower than the normal breathing rate range. We performed an exploratory analysis for those samples when we have high or low breathing rates. We noticed that both algorithms perform poorly with breathing rate that is outside the normal breathing range. We need to adjust the parameters for abnormal breathing range.

9 CONCLUSION

This work proposes a multimodal approach for breathing rate monitoring using earbuds. We use a signal processing-based pipeline that leverages the motion sensor from the earbuds and a machine learning-based pipeline that takes the audio recordings from the earbuds to calculate the breathing rate. Our results suggest that we can estimate the breathing rate using the earbuds with a good accuracy (MAE < 2BPM) on a list of comprehensive tasks in both Lab and Home environments. This work demonstrates the feasibility of using multimodal approach for breathing rate monitoring. We also present the challenges of breathing rate monitoring outside the controlled environment by presenting some empirical evidence. Our work could provide useful direction for designing a useful breathing monitoring system at home environment and could be extended towards a continuous breathing monitoring solution.

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REFERENCES

- [1] Fadel Adib, Hongzi Mao, Zachary Kabelac, Dina Katabi, and Robert C. Miller. 2015. Smart Homes That Monitor Breathing and Heart Rate. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (CHI '15). Association for Computing Machinery, New York, NY, USA, 837–846. <https://doi.org/10.1145/2702123.2702200>
- [2] Tousif Ahmed, Md Mahbubur Rahman, Mohsin Yusuf Ahmed, Ebrahim Nemati, Minh Dinh, Nathan Folkman, Jilong Kuang, and Alex Gao. 2021. RRMonitor: A Resource-Aware End-to-End System for Continuous Monitoring of Respiration Rate Using Earbuds. , 2463–2467 pages. <https://doi.org/10.1109/EMBC46164.2021.9631109>
- [3] Heba Aly and Moustafa Youssef. 2016. Zephyr: Ubiquitous accurate multi-sensor fusion-based respiratory rate estimation using smartphones. In *IEEE INFOCOM 2016 - The 35th Annual IEEE International Conference on Computer Communications*. 1–9. <https://doi.org/10.1109/INFOCOM.2016.7524401>
- [4] Davide Anguita, Alessandro Ghio, Luca Oneto, Xavier Parra Perez, and Jorge Luis Reyes Ortiz. 2013. A public domain dataset for human activity recognition using smartphones. In *Proceedings of the 21th international European symposium on artificial neural networks, computational intelligence and machine learning*. 437–442.
- [5] André Bestbier and Pieter R. Fourie. 2018. Development of a vital signs monitoring wireless ear probe. In *2018 3rd Biennial South African Biomedical Engineering Conference (SAIBMEC)*. 1–5. <https://doi.org/10.1109/SAIBMEC.2018.8363196>
- [6] Jason Bolland, Elaine Bolland, and David Brooks. 2013. Importance of the correct diagnosis of opioid-induced respiratory depression in adult cancer patients and titration of naloxone. *Clinical medicine (London, England)* 13 (04 2013), 149–51. <https://doi.org/10.7861/clinmedicine.13-2-149>
- [7] Data Bridge. 2021. Global Hearable Market – Industry Trends and Forecast to 2028. Retrieved Aug 2, 2021 from <https://www.databridgemarketresearch.com/reports/global-hearable-market>
- [8] Businesswire. 2020. Global Hearables Market (2019 to 2026) - by Products, Type, Connectivity Technology, and End-user. Retrieved Aug 2, 2021 from <https://www.businesswire.com/news/home/20200818005423/en/Global-Hearables-Market-2019-to-2026---by-Products-Type-Connectivity-Technology-and-End-user---ResearchAndMarkets.com>
- [9] Kayla-Jade Butkow, Ting Dang, Andrea Ferlini, Dong Ma, and Cecilia Mascolo. 2021. Motion-resilient Heart Rate Monitoring with In-ear Microphones. arXiv:2108.09393 [cs.HC]
- [10] Soujanya Chatterjee, Md Mahbubur Rahman, Tousif Ahmed, Nazir Saleheen, Ebrahim Nemati, Viswam Nathan, Korosh Vatanparvar, and Jilong Kuang. 2020. Assessing Severity of Pulmonary Obstruction from Respiration Phase-Based Wheeze-Sensing Using Mobile Sensors. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376444>
- [11] Romit Roy Choudhury. 2021. Earable Computing: A New Area to Think About. In *Proceedings of the 22nd International Workshop on Mobile Computing Systems and Applications* (Virtual, United Kingdom) (HotMobile '21). Association for Computing Machinery, New York, NY, USA, 147–153. <https://doi.org/10.1145/3446382.3450216>
- [12] Mary E Crocker, Shakir Hossen, Dina Goodman, Suzanne M Sinkovich, Miles Kirby, Lisa M Thompson, Ghislaine Rosa, Sarada S Garg, Gurusamy Thangavel, Eric D McCollum, Jennifer Peel, Thomas Clasen, and William Checkley. 2020. Effects of high altitude on respiratory rate and oxygen saturation reference values in healthy infants and children younger than 2 years in four countries: a cross-sectional study. *The Lancet Global Health* 8 (3 2020), e362–e373. Issue 3. [https://doi.org/10.1016/S2214-109X\(19\)30543-1](https://doi.org/10.1016/S2214-109X(19)30543-1)
- [13] Consumer Technology Association (CTA). 2022. Performance Criteria and Testing Protocols for Breathing Parameters (CTA-2102). Retrieved Dec 2, 2022 from <https://shop.cta.tech/collections/standards/products/performance-criteria-and-testing-protocols-for-breathing-parameters-cta-2102>
- [14] Ruixuan Dai, Chenyang Lu, Michael Avidan, and Thomas Kannampallil. 2021. RespWatch: Robust Measurement of Respiratory Rate on Smartwatches with Photoplethysmography. In *Proceedings of the International Conference on Internet-of-Things Design and Implementation* (Charlottesville, VA, USA) (IoTDI '21). Association for Computing Machinery, New York, NY, USA, 208–220. <https://doi.org/10.1145/3450268.3453531>
- [15] Tin T. Dang, Khai Q. Nguyen Tran, Hoang M. Le, Khoa Tran, and Anh Dinh. 2018. A Tool Using Ultrasonic Sensor for Measuring Breathing Rate. In *6th International Conference on the Development of Biomedical Engineering in Vietnam (BME6)*, Toi Vo Van, Thanh An Nguyen Le, and Thang Nguyen Duc (Eds.). Springer Singapore, Singapore, 139–143.
- [16] Chendan Dou and Hao Huan. 2021. Full Respiration Rate Monitoring Exploiting Doppler Information with Commodity Wi-Fi Devices. *Sensors* 21, 10 (2021). <https://www.mdpi.com/1424-8220/21/10/3505>
- [17] John F Fieselmann, Michael S Hendryx, Charles M Helms, and Douglas S Wakefield. 1993. Respiratory rate predicts cardiopulmonary arrest for internal medicine inpatients. *Journal of General Internal Medicine* 8 (1993), 354–360. Issue 7. <https://doi.org/10.1007/BF02600071>
- [18] Tracy Flenady, Trudy Dwyer, and Judith Applegarth. 2017. Accurate respiratory rates count: So should you! , 45–47 pages. Issue 1. <https://doi.org/10.1016/j.aenj.2016.12.003>
- [19] Yang Gao, Wei Wang, Vir V. Phoha, Wei Sun, and Zhanpeng Jin. 2019. EarEcho: Using Ear Canal Echo for Wearable Authentication. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 3, 3, Article 81 (sep 2019), 24 pages. <https://doi.org/10.1145/3351239>
- [20] Shkurta Gashi, Aaqib Saeed, Alessandra Vicini, Elena Di Lascio, and Silvia Santini. 2021. Hierarchical Classification and Transfer Learning to Recognize Head Gestures and Facial Expressions Using Earbuds. In *Proceedings of the 2021 International Conference on Multimodal Interaction* (Montréal, QC, Canada) (ICMI '21). Association for Computing Machinery, New York, NY, USA, 168–176. <https://doi.org/10.1145/3462244.3479921>
- [21] Valentin Goverdovsky, Wilhelm von Rosenberg, Takashi Nakamura, David Looney, David Sharp, C. Papavassiliou, Mary Morrell, and Danilo Mandic. 2016. Hearables: Multimodal physiological in-ear sensing. *Scientific Reports* 7 (09 2016). <https://doi.org/10.1038/s41598-017-06925-2>
- [22] Mariel Grassmann, Elke Vlemincx, Andreas Von Leupoldt, Justin M. Mittelstädt, and Omer Van Den Bergh. 2016. Respiratory changes in response to cognitive load: A systematic review. <https://doi.org/10.1155/2016/8146809>
- [23] Yu Gu, Xiang Zhang, Zhi Liu, and Fuji Ren. 2019. WiFi-Based Real-Time Breathing and Heart Rate Monitoring during Sleep. In *2019 IEEE Global Communications Conference (GLOBECOM)*. 1–6. <https://doi.org/10.1109/GLOBECOM38437.2019.9014297>
- [24] Migyeong Gwak, Korosh Vatanparvar, Jilong Kuang, and Alex Gao. 2022. Motion-Based Respiratory Rate Estimation with Motion Artifact Removal Using Video of Face and Upper Body. In *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. 1961–1967. <https://doi.org/10.1109/EMBC48229.2022.9871231>
- [25] Unsoo Ha, Sohrab Madani, and Fadel Adib. 2021. WiStress: Contactless Stress Monitoring Using Wireless Signals. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 5, 3, Article 103 (Sept. 2021), 37 pages. <https://doi.org/10.1145/3478121>
- [26] M. A. Hall. 1998. *Correlation-based Feature Subset Selection for Machine Learning*. Ph.D. Dissertation. University of Waikato, Hamilton, New Zealand.
- [27] Tian Hao, Chongguang Bi, Guoliang Xing, Roxane Chan, and Linlin Tu. 2017. MindfulWatch: A Smartwatch-Based System For Real-Time Respiration Monitoring During Meditation. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 1, 3, Article 57 (Sept. 2017), 19 pages. <https://doi.org/10.1145/3130922>
- [28] Javier Hernandez, Yin Li, James M. Rehg, and Rosalind W. Picard. 2014. BioGlass: Physiological parameter estimation using a head-mounted wearable device. In *2014 4th International Conference on Wireless Mobile Communication and Healthcare - Transforming Healthcare Through Innovations in Mobile and Wireless Technologies (MOBIHEALTH)*. 55–58. <https://doi.org/10.1109/MOBIHEALTH.2014.7015908>
- [29] Javier Hernandez, Daniel McDuff, and Rosalind W. Picard. 2015. Biowatch: estimation of heart and breathing rates from wrist motions. In *2015 9th International Conference on Pervasive Computing Technologies for Healthcare (PervasiveHealth)*. IEEE, 169–176.
- [30] Nick Hunn. 2014. Hearables – the new Wearables. Retrieved Oct 15, 2021 from <https://www.wearable-technologies.com/2014/04/hearables-the-new-wearables/>
- [31] Bashima Islam, Md Mahbubur Rahman, Tousif Ahmed, Mohsin Yusuf Ahmed, Md Mehedi Hasan, Viswam Nathan, Korosh Vatanparvar, Ebrahim Nemati, Jilong Kuang, and Jun Alex Gao. 2021. BreathTrack: Detecting Regular Breathing Phases from Unannotated Acoustic Data Captured by a Smartphone. 5, 3, Article 124 (Sept. 2021), 22 pages. <https://doi.org/10.1145/3478123>

- [32] Yincheng Jin, Yang Gao, Xiaotao Guo, Jun Wen, Zhengxiong Li, and Zhanpeng Jin. 2022. EarHealth: An Earphone-Based Acoustic Oscope for Detection of Multiple Ear Diseases in Daily Life. In *Proceedings of the 20th Annual International Conference on Mobile Systems, Applications and Services* (Portland, Oregon) (*MobiSys '22*). Association for Computing Machinery, New York, NY, USA, 397–408. <https://doi.org/10.1145/3498361.3538935>
- [33] Wall Street Journal. 2021. Apple Studying Potential of AirPods as Health Device. Retrieved Oct 15, 2021 from <https://www.wsj.com/articles/apple-studying-potential-of-airpods-as-health-device-11634122800>
- [34] Carol Kelly. 2018. Respiratory rate 1: why measurement and recording are crucial. *Nursing Mirror* 114, 4 (2 April 2018), 23–24.
- [35] Jochen Kempfle and Kristof Van Laerhoven. 2018. Respiration Rate Estimation with Depth Cameras: An Evaluation of Parameters. In *Proceedings of the 5th International Workshop on Sensor-Based Activity Recognition and Interaction* (Berlin, Germany) (*iWOAR '18*). Association for Computing Machinery, New York, NY, USA, Article 4, 10 pages. <https://doi.org/10.1145/3266157.3266208>
- [36] Diederik P Kingma and Jimmy Ba. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).
- [37] Agni Kumar, Vikramjit Mitra, Carolyn Oliver, Adeeti Ullal, Matt Biddulph, and Irida Mance. 2021. Estimating Respiratory Rate From Breath Audio Obtained Through Wearable Microphones. <https://arxiv.org/pdf/2107.14028.pdf>
- [38] Bishal Lamichhane, Ebrahim Nemati, Tousif Ahmed, Mahbubur Rahman, Jilong Kuang, and Alex Gao. 2022. A Template Matching Based Cough Detection Algorithm Using IMU Data From Earbuds. In *2022 IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI)*. 01–04. <https://doi.org/10.1109/BHI56158.2022.9926839>
- [39] Daniyal Liaqat, Mohamed Abdalla, Pegah Abed-Esfahani, Moshe Gabel, Tatiana Son, Robert Wu, Andrea Gershon, Frank Rudzicz, and Eyal De Lara. 2019. Wear-Breathing: Real World Respiratory Rate Monitoring Using Smartwatches. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 3, 2, Article 56 (June 2019), 22 pages. <https://doi.org/10.1145/3328927>
- [40] PneumaCare LIMITED. 2016. Thora-3Di Model T-01 510(k) Premarket Notification. Retrieved Dec 2, 2022 from <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm?ID=K151940>
- [41] Haipeng Liu, John Allen, Dingchang Zheng, and Fei Chen. 2019. Recent development of respiratory rate measurement technologies. *Physiological Measurement* 40, 7 (aug 2019), 07TR01. <https://doi.org/10.1088/1361-6579/ab299e>
- [42] Deborah Lupton. 2016. *The quantified self*. John Wiley & Sons.
- [43] Alexis Martin and Jérémie Voix. 2018. In-Ear Audio Wearable: Measurement of Heart and Breathing Rates for Health and Safety Monitoring. *IEEE Transactions on Biomedical Engineering* 65, 6 (2018), 1256–1263. <https://doi.org/10.1109/TBME.2017.2720463>
- [44] Yuri Masaoka and Ikuo Homma. 1997. Anxiety and respiratory patterns: their relationship during mental stress and physical load. *International Journal of Psychophysiology* 27, 2 (1997), 153–159.
- [45] Yuri Masaoka and Ikuo Homma. 1997. Anxiety and respiratory patterns: their relationship during mental stress and physical load. , 153159 pages.
- [46] Carlo Massaroni, Andrea Nicolò, Emiliano Schena, and Massimo Sacchetti. 2020. Remote Respiratory Monitoring in the Time of COVID-19. *Frontiers in physiology* 11 (2020), 635. <https://doi.org/10.3389/fphys.2020.00635>
- [47] Se Dong Min, Jin Kwon Kim, Hang Sik Shin, Yong Hyeon Yun, Chung Keun Lee, and Myoungcho Lee. 2010. Noncontact Respiration Rate Measurement System Using an Ultrasonic Proximity Sensor. *IEEE Sensors Journal* 10, 11 (2010), 1732–1739. <https://doi.org/10.1109/JSEN.2010.2044239>
- [48] Yunyoung Nam, Youngsun Kong, Bersain Reyes, Natasa Reljin, and Ki H. Chon. 2016. Monitoring of heart and breathing rates using dual cameras on a smartphone. *PLoS ONE* 11 (3 2016). Issue 3. <https://doi.org/10.1371/journal.pone.0151013>
- [49] Yunyoung Nam, Bersain A. Reyes, and Ki H. Chon. 2016. Estimation of Respiratory Rates Using the Built-in Microphone of a Smartphone or Headset. *IEEE Journal of Biomedical and Health Informatics* 20, 6 (2016), 1493–1501. <https://doi.org/10.1109/JBHI.2015.2480838>
- [50] Aravind Natarajan, Hao Wei Su, Conor Heneghan, Leanna Blunt, Corey O'Connor, and Logan Niehaus. 2021. Measurement of respiratory rate using wearable devices and applications to COVID-19 detection. *npj Digital Medicine* 4 (12 2021). Issue 1. <https://doi.org/10.1038/s41746-021-00493-6>
- [51] Colver Ken Howe Ne, Jameel Muzaffar, Aakash Amlani, and Manohar Bance. 2021. Hearables, in-ear sensing devices for bio-signal acquisition: a narrative review. *Expert Review of Medical Devices* 18, sup1 (2021), 95–128. <https://doi.org/10.1080/17434440.2021.2014321> PMID: 34904507.
- [52] Ebrahim Nemati, Shibo Zhang, Tousif Ahmed, Md. Mahbubur Rahman, Jilong Kuang, and Alex Gao. 2021. CoughBuddy: Multi-Modal Cough Event Detection Using Earbuds Platform. In *2021 IEEE 17th International Conference on Wearable and Implantable Body Sensor Networks (BSN)*. 1–4. <https://doi.org/10.1109/BSN51625.2021.9507017>
- [53] Andrea Nicolò, Samuele M. Marcora, and Massimo Sacchetti. 2016. Respiratory frequency is strongly associated with perceived exertion during time trials of different duration. *Journal of Sports Sciences* 34, 13 (2016), 1199–1206. <https://doi.org/10.1080/02640414.2015.1102315> PMID: 26503587.
- [54] Andrea Nicolò, Carlo Massaroni, and Louis Passfield. 2017. Respiratory Frequency during Exercise: The Neglected Physiological Measure. *Frontiers in Physiology* 8 (2017), 922. <https://doi.org/10.3389/fphys.2017.00922>
- [55] Andrea Nicolò, Carlo Massaroni, Emiliano Schena, and Massimo Sacchetti. 2020. The Importance of Respiratory Rate Monitoring: From Healthcare to Sport and Exercise. *Sensors (Basel, Switzerland)* 20, 21 (Nov. 2020). <https://doi.org/10.3390/s20216396>
- [56] Jon Nordby. 2019. emlearn: Machine Learning inference engine for Microcontrollers and Embedded Devices. <https://doi.org/10.5281/zenodo.2589394>
- [57] Royal College of Physicians. 2017. National early warning score (NEWS): standardising the assessment of acute-illness severity in the NHS—report of a working party. Retrieved Oct 15, 2021 from <https://www.rcplondon.ac.uk/projects/outputs/national-early-warning-score-news-2>
- [58] Med Device Online. 2021. Meet Hearables, The Next Revolutionary Medical Devices. Retrieved Oct 15, 2021 from <https://www.meddeviceonline.com/doc/meet-hearables-the-next-revolutionary-medical-devices-0001>
- [59] Aleš Procházka, Martin Schätz, Oldřich Vyšata, and Martin Vališ. 2016. Microsoft Kinect Visual and Depth Sensors for Breathing and Heart Rate Analysis. *Sensors* 16, 7 (2016). <https://www.mdpi.com/1424-8220/16/7/996>
- [60] Md Mahbubur Rahman, Mohsin Yusuf Ahmed, Tousif Ahmed, Bashima Islam, Viswam Nathan, Korosh Vatanparvar, Ebrahim Nemati, Daniel McCaffrey, Jilong Kuang, and Jun Alex Gao. 2020. BreathEasy: Assessing Respiratory Diseases Using Mobile Multimodal Sensors. In *Proceedings of the 2020 International Conference on Multimodal Interaction* (Virtual Event, Netherlands) (*ICMI '20*). Association for Computing Machinery, New York, NY, USA, 41–49. <https://doi.org/10.1145/3382507.3418852>
- [61] Md Mahbubur Rahman, Tousif Ahmed, Mohsin Yusuf Ahmed, Minh Dinh, Ebrahim Nemati, Jilong Kuang, and Jun Alex Gao. 2022. BreatheBuddy: Tracking Real-Time Breathing Exercises for Automated Biofeedback Using Commodity Earbuds. *Proc. ACM Hum.-Comput. Interact.* 6, MHCI, Article 213 (sep 2022), 18 pages. <https://doi.org/10.1145/3546748>
- [62] Md Mahbubur Rahman, Tousif Ahmed, Mohsin Yusuf Ahmed, Ebrahim Nemati, Minh Dinh, Nathan Folkman, Md Mehedi Hasan, Jilong Kuang, and Jun Alex Gao. 2021. Towards Motion-Aware Passive Resting Respiratory Rate Monitoring Using Earbuds. In *2021 IEEE 17th International Conference on Wearable and Implantable Body Sensor Networks (BSN)*. 1–4. <https://doi.org/10.1109/BSN51625.2021.9507016>
- [63] Md. Mahbubur Rahman, Ebrahim Nemati, Viswam Nathan, and Jilong Kuang. 2020. InstantRR: Instantaneous Respiratory Rate Estimation on Context-Aware Mobile Devices. In *13th EAI International Conference on Body Area Networks*, Chika Sugimoto, Hamed Farhadi, and Matti Hämäläinen (Eds.). Springer International Publishing, Cham, 267–281.
- [64] Tauhidur Rahman, Alexander T. Adams, Mi Zhang, Erin Cherry, Bobby Zhou, Huaishu Peng, and Tanzeem Choudhury. 2014. BodyBeat: A Mobile System for Sensing Non-Speech Body Sounds (*MobiSys '14*). Association for Computing Machinery, New York, NY, USA, 2–13. <https://doi.org/10.1145/2594368.2594386>
- [65] Ruth Ravichandran, Elliot Saba, Ke-Yu Chen, Mayank Goel, Sidhant Gupta, and Shwetak N. Patel. 2015. WiBreathe: Estimating respiration rate using wireless signals in natural settings in the home. In *2015 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. 131–139. <https://doi.org/10.1109/PERCOM.2015.7146519>
- [66] Tobias Röddiger, Christopher Clarke, Paula Breitling, Tim Schneegans, Haibin Zhao, Hans Gellersen, and Michael Beigl. 2022. Sensing with Earables: A Systematic Literature Review and Taxonomy of Phenomena. 6, 3, Article 135 (sep 2022), 57 pages. <https://doi.org/10.1145/3550314>
- [67] Tobias Röddiger, Daniel Wolfram, David Laubenstein, Matthias Budde, and Michael Beigl. 2019. Towards Respiration Rate Monitoring Using an In-Ear Headphone Inertial Measurement Unit. In *Proceedings of the 1st International Workshop on Earable Computing* (London, United Kingdom) (*EarComp '19*). Association for Computing Machinery, New York, NY, USA, 48–53. <https://doi.org/10.1145/3345615.3361130>
- [68] Sandy Rolfe. 2019. The importance of respiratory rate monitoring. *British journal of nursing (Mark Allen Publishing)* 28, 8 (April 2019), 504–508. <https://doi.org/10.12968/bjon.2019.28.8.504> Place: England.
- [69] Chiara Romano, Emiliano Schena, Sergio Silvestri, and Carlo Massaroni. 2021. Non-contact respiratory monitoring using an RGB camera for real-world applications. *Sensors* 21 (8 2021). Issue 15. <https://doi.org/10.3390/s21155126>
- [70] J Brady Scott and Ramandeep Kaur. 2020. Monitoring Breathing Frequency, Pattern, and Effort. *Respiratory Care* 65, 6 (2020), 793–806. <https://doi.org/10.4187/respcare.07439> arXiv: <http://rc.rcjournal.com/content/65/6/793.full.pdf>
- [71] D.M. Seath and G.D. Miller. 1946. The Relative Importance of High Temperature and High Humidity as Factors Influencing Respiration Rate, Body Temperature, and Pulse Rate of Dairy Cows. *Journal of Dairy Science* 29, 7 (1946), 465–472. [https://doi.org/10.3168/jds.S0022-0302\(46\)92502-7](https://doi.org/10.3168/jds.S0022-0302(46)92502-7)
- [72] Ville-Pekka Seppä, Anna S. Pelkonen, Anne Kotaniemi-Syrjänen, Jari Viik, Mika J. Mäkelä, and L. Pekka Malmberg. 2016. Tidal flow variability measured by impedance pneumography relates to childhood asthma risk. *European Respiratory Journal* 47, 6 (2016), 1687–1696. <https://doi.org/10.1183/13993003.00989-2015>

- arXiv:<https://erj.ersjournals.com/content/47/6/1687.full.pdf>
- [73] Pragya Sharma, Xiaonan Hui, Jianlin Zhou, Thomas B Conroy, and Edwin C Kan. 2020. Wearable radio-frequency sensing of respiratory rate, respiratory volume, and heart rate. *npj Digital Medicine* 3 (2020), 98. Issue 1. <https://doi.org/10.1038/s41746-020-0307-6>
 - [74] William M Suess, A Barney Alexander, Deborah D Smith, Helga W Sweeney, and Richard J Marion. 1980. The effects of psychological stress on respiration: a preliminary study of anxiety and hyperventilation. *Psychophysiology* 17, 6 (1980), 535–540.
 - [75] Xiao Sun, Li Qiu, Yibo Wu, Yeming Tang, and Guohong Cao. 2017. SleepMonitor: Monitoring Respiratory Rate and Body Position During Sleep Using Smartwatch. *Proc. ACM Interact. Mob. Wearable Ubiquitous Tech.* 1, 3, Article 104 (Sept. 2017), 22 pages. <https://doi.org/10.1145/3130969>
 - [76] Zephyr Technology. 2012. BioHarness 3.0. User Manual. Retrieved Aug 2, 2021 from <https://www.zephyranywhere.com/media/download/bioharness3-user-manual.pdf>
 - [77] Michael J. Tipton, Abbi Harper, Julian F. R. Paton, and Joseph T. Costello. 2017. The human ventilatory response to stress: rate or depth? *The Journal of Physiology* 595, 17 (2017), 5729–5752. <https://doi.org/10.1113/JP274596>
 - [78] Korosh Vatanparvar, Migyeong Gwak, Li Zhu, Jilong Kuang, and Alex Gao. 2022. Respiration Rate Estimation from Remote PPG via Camera in Presence of Non-Voluntary Artifacts. In *2022 IEEE-EMBS International Conference on Wearable and Implantable Body Sensor Networks (BSN)*. 1–4. <https://doi.org/10.1109/BSN56160.2022.9928485>
 - [79] Elke Vlemincx, Joachim Taelman, Steven De Puter, Ilse Van Diest, and Omer Van Den Bergh. 2011. Sigh rate and respiratory variability during mental load and sustained attention. *Psychophysiology* 48, 1 (2011), 117–120.
 - [80] Sylvain Wallez. 2021 (accessed Jan, 2021). *Savitzky Golay Filter*. <https://github.com/swallez/savitzky-golay-filter>.
 - [81] Xuyu Wang, Runze Huang, Chao Yang, and Shiwen Mao. 2021. Smartphone Sonar-Based Contact-Free Respiration Rate Monitoring. *ACM Trans. Comput. Healthcare* 2, 2, Article 15 (feb 2021), 26 pages. <https://doi.org/10.1145/3436822>
 - [82] Yuntao Wang, Xiyuxing Zhang, Jay M. Chakalasiya, Xuhai Xu, Yu Jiang, Yuang Li, Shwetak Patel, and Yuanchun Shi. 2022. HearCough: Enabling continuous cough event detection on edge computing hearables. *Methods* 205 (2022), 53–62. <https://doi.org/10.1016/j.ymeth.2022.05.002>
 - [83] Zihan Wang, Tousif Ahmed, Md. Mahbubur Rahman, Mohsin Y Ahmed, Ebrahim Nemati, Jilong Kuang, and Alex Gao. 2022. Real-Time Breathing Phase Detection Using Earbuds Microphone. In *2022 IEEE-EMBS International Conference on Wearable and Implantable Body Sensor Networks (BSN)*. 1–4. <https://doi.org/10.1109/BSN56160.2022.9928520>
 - [84] Stephen E Widen, Sara Bäsjo, Claes Möller, and Kim Kähäri. 2017. Headphone listening habits and hearing thresholds in swedish adolescents. *Noise & health* 19, 88 (2017), 125–132. https://doi.org/10.4103/nah.NAH_65_16
 - [85] Devy Widjaja, Michele Orini, Elke Vlemincx, and Sabine Van Huffel. 2013. Cardiorespiratory dynamic response to mental stress: A multivariate time-frequency analysis. *Computational and Mathematical Methods in Medicine* 2013 (2013). <https://doi.org/10.1155/2013/451857>
 - [86] Xuhai Xu, Haitian Shi, Xin Yi, WenJia Liu, Yukang Yan, Yuanchun Shi, Alex Mariakakis, Jennifer Mankoff, and Anind K. Dey. 2020. EarBuddy: Enabling On-Face Interaction via Wireless Earbuds. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376836>
 - [87] Aina M. Yañez, Dolores Guerrero, Rigoberto Pérez de Alejo, Francisco García-Rio, Jose Luis Alvarez-Sala, Miriam Calle-Rubio, Rosa Malo de Molina, Manuel Valle Falcones, Piedad Ussetti, Jaume Sauleda, Enrique Zamora García, Jose Miguel Rodríguez-González-Moro, Mercedes Franco Gay, Maties Torrent, and Alvar Agustí. 2012. Monitoring Breathing Rate at Home Allows Early Identification of COPD Exacerbations. *Chest* 142, 6 (2012), 1524–1529. <https://doi.org/10.1378/chest.11-2728>
 - [88] Shibo Zhang, Ebrahim Nemati, Minh Dinh, Nathan Folkman, Tousif Ahmed, Mahbubur Rahman, Jilong Kuang, Nabil Alshurafa, and Alex Gao. 2022. Coughtrigger: Earbuds IMU Based Cough Detection Activator Using An Energy-Efficient Sensitivity-Prioritized Time Series Classifier. In *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 1–5. <https://doi.org/10.1109/ICASSP43922.2022.9746334>
 - [89] Fang Zhao, Meng Li, Yi Qian, and Joe Z. Tsien. 2013. Remote Measurements of Heart and Respiration Rates for Telemedicine. *PLOS ONE* 8, 10 (10 2013), 1–14. <https://doi.org/10.1371/journal.pone.0071384>

A HEAD MOTION INSTRUCTIONS

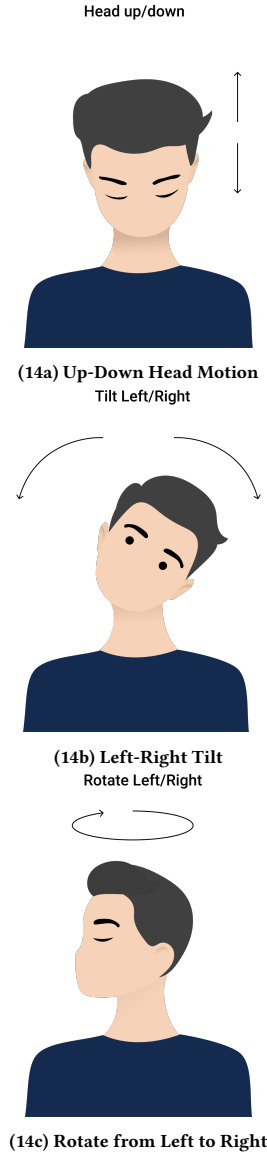


Figure 14: Head Motion Task. The participants performed each head movement for 30 seconds.

B ANNOTATION PORTAL

We designed an annotation portal for annotation purposes. In our annotation portal, an annotator can upload both the time-series

data and audio data. Figure 15, ??, 16, 17, and 18 shows some snapshot of our annotation portal. Using the annotation portal, the annotator can upload a list of files prepared for annotations. Some brief details about different annotations are given below:

B.1 Chestband Annotation

Figure 15 and ?? shows two sample snapshots of chestband annotation. We use a semi-automated process for chestband annotation, where we run a peak-detection algorithm on the breathing waveform collected from chestband. However, the automated algorithm sometimes misses peaks and valleys (Figure ??). The human annotator reviews all the annotations and adds peaks and valleys. We added a short key for assisting the annotator. On the annotation portal, the annotator can write notes and mark the file as 'Hidden' to imply that the annotation is not annotatable.

B.2 Audio Annotation

Figure 16 shows a snapshot of our audio annotation portal. The annotator can upload audio and visualize the waveform and spectrogram in the portal. The annotator can listen to the audio for annotation and pause the audio when required. The spectrogram annotation is designed to facilitate the breathing signal annotation.

B.3 Head Motion Annotation

For developing the head motion classifier, the annotators also annotated the motion sensor data and annotated it using the portal. Figure 17 shows an example of head motion annotation. We use gyroscope data for annotation, so when there is significant movement, the time-series representation of the signal indicates the movement period. The annotator annotated the regions in the signal by visualizing the gyroscope data in the portal.

B.4 Multimodal Annotation Review

In the annotation portal, the annotators can upload multiple time series data to check the quality of the signal. For example, when the chestband fails to capture the breathing motion, we use the audio data to annotate the breathing signal. Figure 18 shows an example of a multimodal annotation portal. In this case, the annotator uploaded audio data, chestband signal, and motion sensor data to confirm the breathing event. As can be seen in the middle part of the image, the chestband failed to capture the breathing signal due to body movements (motion captured in the 3rd part of the figure). However, the audio data seems clean, and we can utilize the audio signal for breathing rate calculation. On such occasions, the annotator marked the chestband quality as bad, and we used audio data for breathing rate calculation (ground truth).

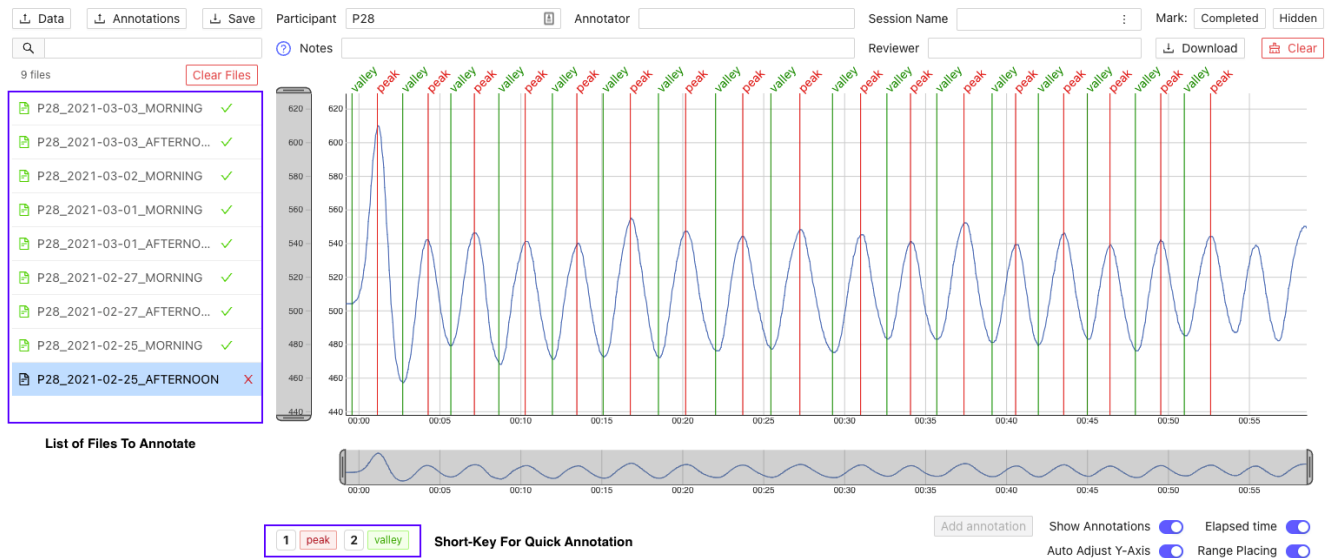
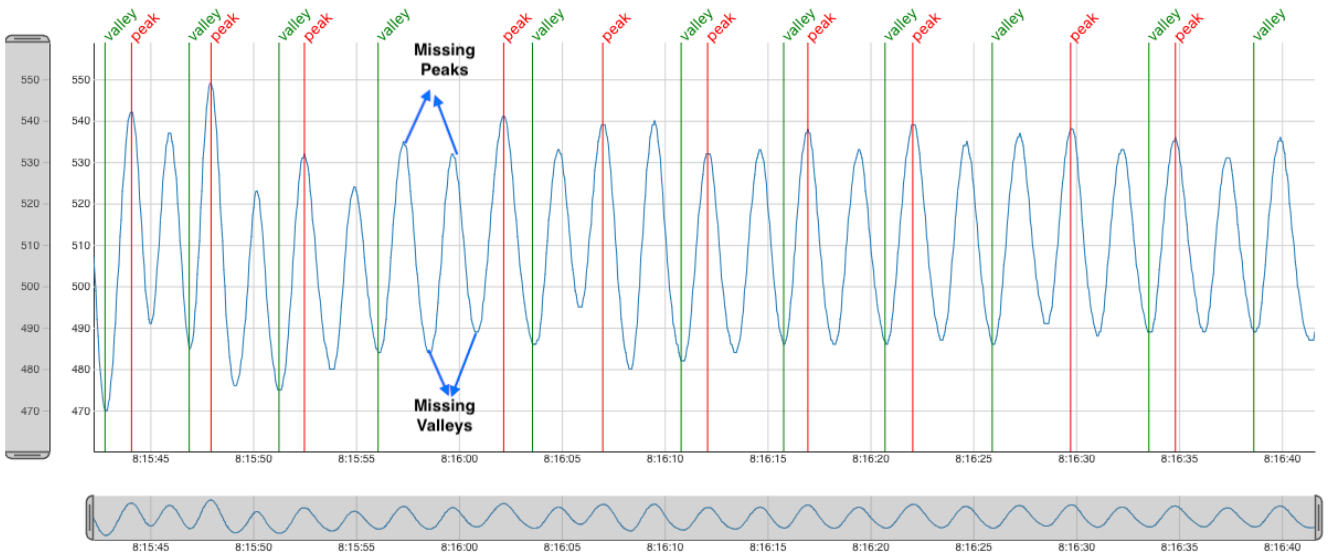


Figure 15: Our designed annotation portal



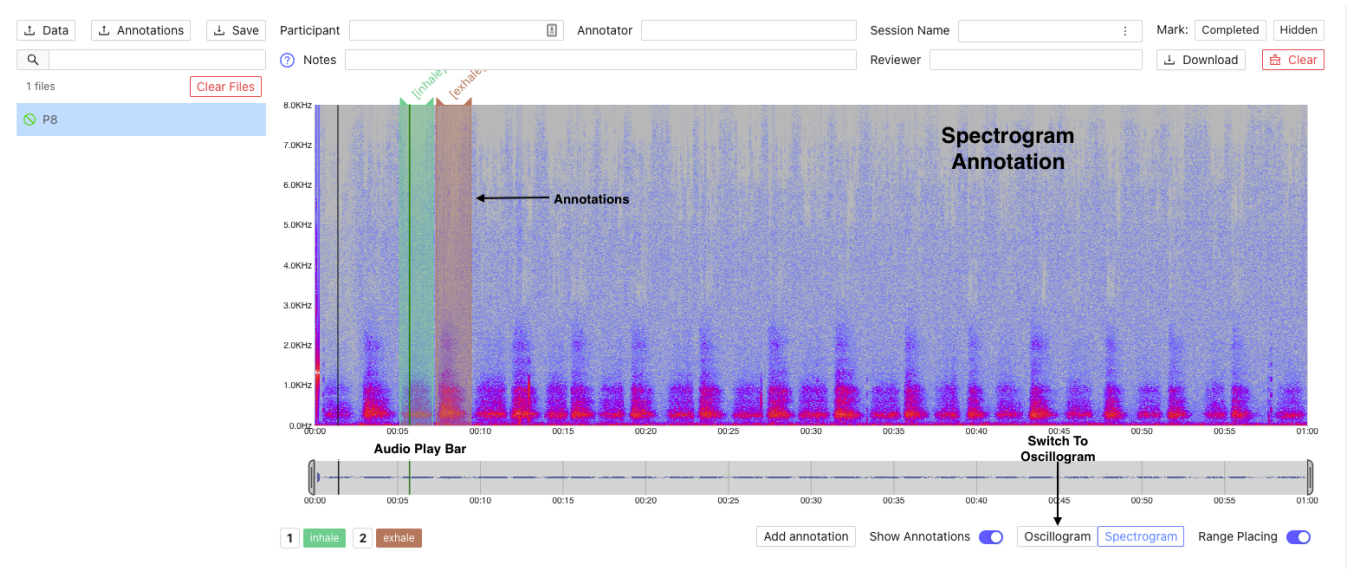


Figure 16: Annotation Portal showing Spectrogram Signal. We mainly annotated Inhale, Exhale, and Noise from the audio data.

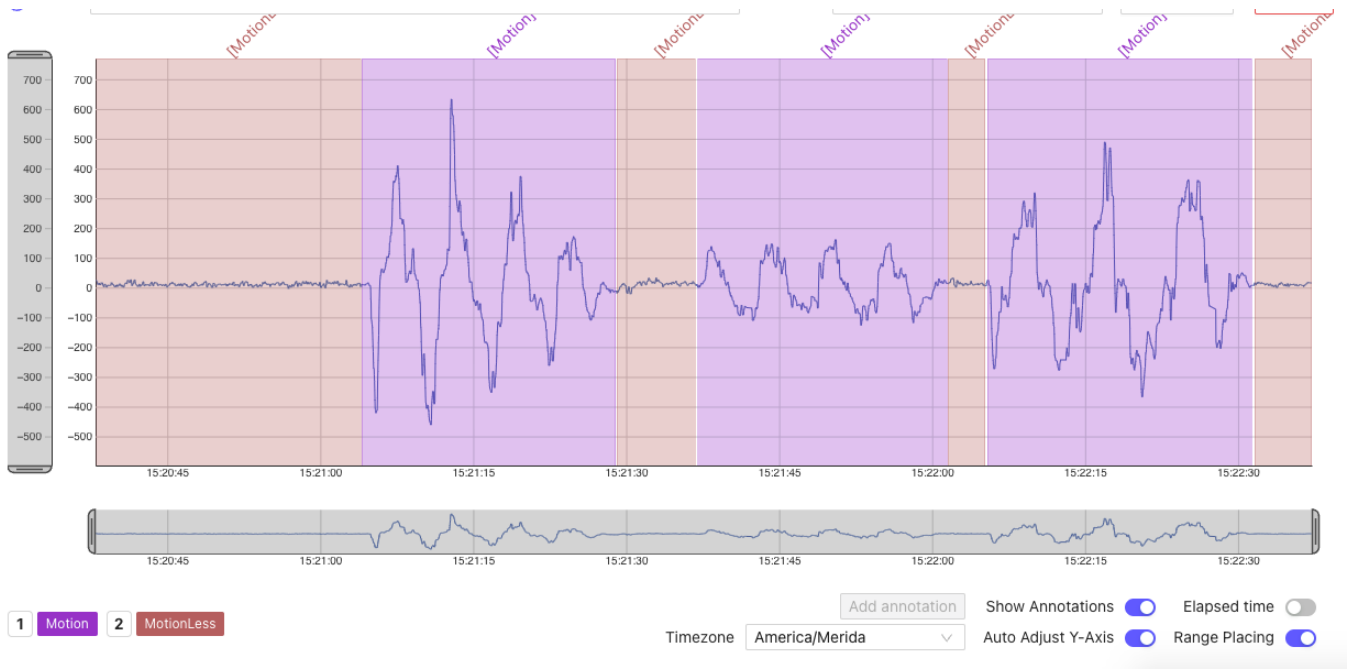


Figure 17: Head Motion Annotation using the annotation portal. We mainly annotated Motion vs Motionless on the Gyroscope signal.



Figure 18: Annotation Review using the annotation portal. It can show multiple signals at the same time.